

Project Management CRM

Project Overview:

The Project Management CRM is developed in Salesforce to streamline project planning, tracking, and team collaboration. It enables organizations to manage projects, assign tasks, monitor progress, track deadlines, and ensure smooth communication among stakeholders. The CRM helps improve productivity, resource allocation, and reporting by automating key processes and maintaining centralized project data.

Objectives:

The main objectives of this CRM are:

- To organize and track multiple projects efficiently
- To manage tasks, milestones, and deadlines
- To improve collaboration between project managers, team members, and clients
- To automate project workflows and approvals
- To generate analytical reports for decision-making

This CRM delivers business value through improved productivity, reduced delays, and better project visibility.

Phase 1 — Problem Understanding & Industry Analysis:

Requirement Gathering

- Need for centralized project & task tracking
- Difficulty in monitoring deadlines & status updates
- Manual reporting led to time delays

Stakeholder Analysis

- Project Manager
- Team Members
- Client / End User

Business Process Mapping

- Project Creation → Task Assignment → Progress Tracking → Status Review

Industry Use Case

Applicable in:

- IT Services
- Consulting
- Construction & Operations
- Startup project teams

AppExchange Exploration

Reviewed:

- Project Management Apps
- Resource Tracking Tools

Selected custom CRM to fit use-case.

Phase 2 — Org Setup & Configuration:

- Company Profile

Defines organization details such as company name, address, language, currency, time zone, and locale used across the Salesforce org.

- Fiscal Year

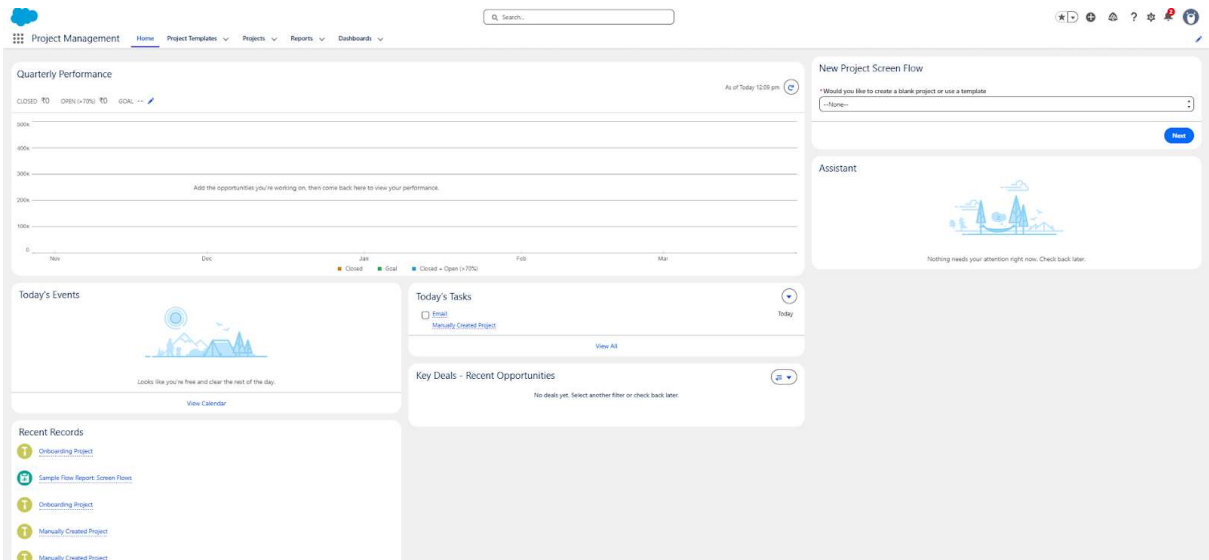
Defines how financial or project reporting periods are structured (standard or custom fiscal year).

- User Roles

Represents the role hierarchy that controls data visibility and record access within the organization.

- Profiles & Permissions

Profiles and permission sets define object access, field permissions, and user capabilities in the CRM.

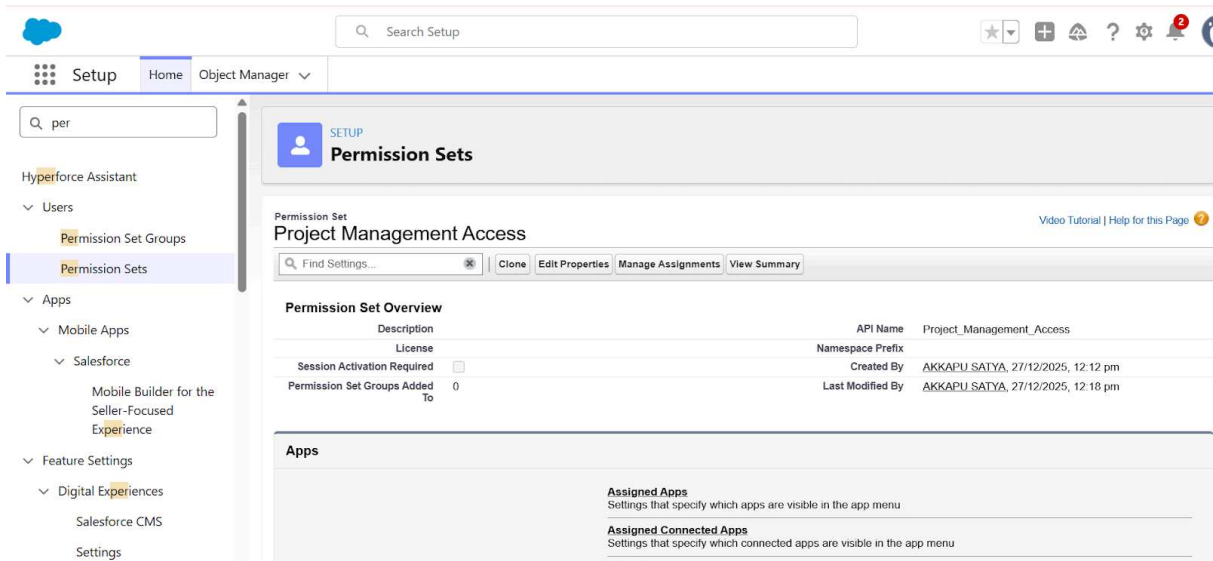


The screenshot shows the Salesforce Projects list view. At the top, there's a navigation bar with 'Project Management' selected, and a search bar. Below the navigation bar, the list view is titled 'Projects' and shows a list of 4 items. The list is sorted by Project Name. The table has columns for Project Name, Status, and Created Date. The projects listed are 'Manually Created Project' and 'Onboarding Project'.

Project Name	Status	Created Date
1 ☐ Manually Created Project		27/12/2025, 8:37 am
2 ☐ Manually Created Project	New	27/12/2025, 9:49 am
3 ☐ Onboarding Project		27/12/2025, 12:07 pm
4 ☐ Onboarding Project		27/12/2025, 10:20 am

The screenshot shows the Salesforce Object Manager setup page. At the top, there's a navigation bar with 'Setup' selected, and a search bar. Below the navigation bar, the page is titled 'Object Manager' and shows a list of 3 items. The list is sorted by Label. The table has columns for Label, API Name, Type, Description, Last Modified, and Deployed. The objects listed are 'Project', 'Project Task Template', and 'Project Template'.

Label	API Name	Type	Description	Last Modified	Deployed
Project	Project__c	Custom Object		27/12/2025	✓
Project Task Template	Project_Task_Template__c	Custom Object		27/12/2025	✓
Project Template	Project_Template__c	Custom Object		27/12/2025	✓



The screenshot shows the Salesforce Setup interface. The left sidebar contains a navigation menu with 'Setup' selected. Under 'Setup', 'Permission Sets' is highlighted. The main content area is titled 'Permission Sets' and shows a specific permission set named 'Project Management Access'. The 'Permission Set Overview' section displays the following details:

Field	Value
Description	Project Management Access
API Name	Project_Management_Access
License	Standard
Session Activation Required	☐
Created By	AKKAPU SATYA, 27/12/2025, 12:12 pm
Last Modified By	AKKAPU SATYA, 27/12/2025, 12:18 pm
Permission Set Groups Added To	0

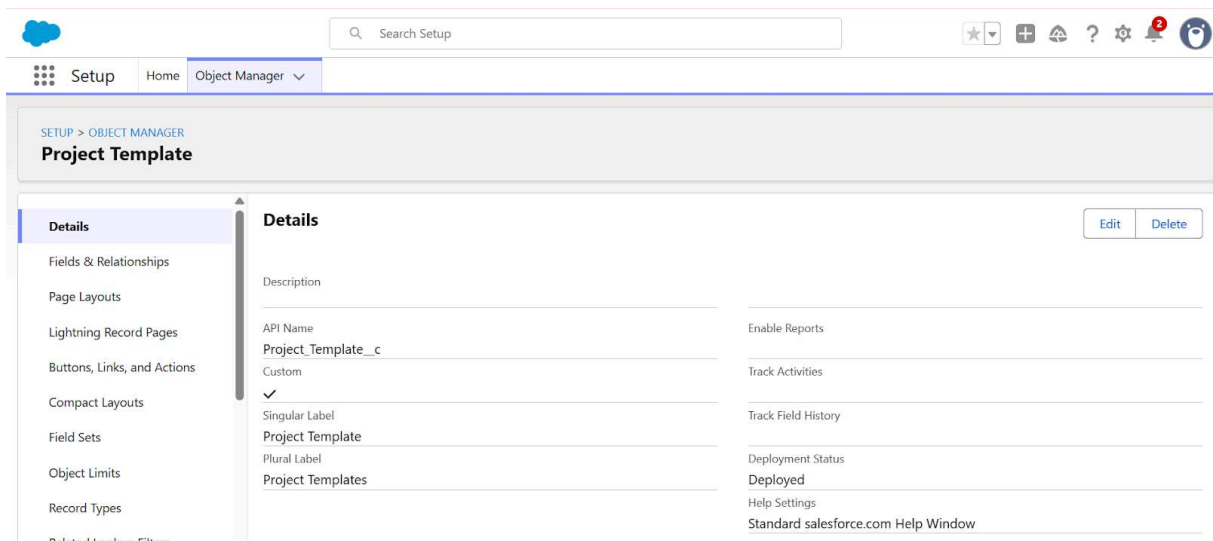
Below the overview, there is a section for 'Apps' with 'Assigned Apps' and 'Assigned Connected Apps'.

Phase 3 — Data Modeling & Relationships:

Standard & Custom Objects Used

Custom Objects:

- Project Template
- Project Task template



The screenshot shows the Salesforce Setup interface for the 'Project Template' custom object. The left sidebar contains a navigation menu with 'Setup' selected. Under 'Setup', 'Object Manager' is selected, and 'Project Template' is highlighted. The main content area is titled 'Project Template' and shows the 'Details' section. The 'Details' section displays the following information:

Field	Value
Description	
API Name	Project_Template__c
Custom	☒
Singular Label	Project Template
Plural Label	Project Templates
Enable Reports	☐
Track Activities	☐
Track Field History	☐
Deployment Status	Deployed
Help Settings	Standard salesforce.com Help Window

Setup

Home

Object Manager

Search Setup

★

+

🔒

?

⚙️

🔔

👤

SETUP > OBJECT MANAGER

Project Task Template

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related List Filters

Details

Description

API Name
Project_Task_Template__c

Custom
✓

Singular Label
Project Task Template

Plural Label
Project Task Templates

Edit

Delete

Enable Reports

Track Activities

Track Field History

Deployment Status
Deployed

Help Settings
Standard salesforce.com Help Window

Fields Used

- Project
- Description
- Due Date Offset

Setup

Home

Object Manager

Search Setup

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👤

SETUP > OBJECT MANAGER

Project Task Template

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related List Filters

Project Task Template Custom Field

Project

Back to Project Task Template

Validation Rules [0]

Custom Field Definition Detail

Edit

Set Field-Level Security

View Field Accessibility

Where is this used?

Field Information

Field Label	Project	Object Name	Project_Task_Template
Field Name	Project	Data Type	Master-Detail
API Name	Project__c		
Description			
Help Text			
Data Owner			
Field Usage			
Data Sensitivity Level			
Compliance Categorization			
Created By	AKKAPU SATYA, 27/12/2025, 8:07 am	Modified By	AKKAPU SATYA, 27/12/2025, 8:07 am

The screenshot shows the Salesforce Setup interface for the 'Project Task Template' custom field. The left sidebar contains a navigation menu with options: Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, and Related Lookup Fields. The main content area is titled 'Project Task Template Custom Field Description' and includes a 'Back to Project Task Template' link. Below this is the 'Custom Field Definition Detail' section with tabs for 'Edit', 'Set Field-Level Security', 'View Field Accessibility', and 'Where is this used?'. The 'Field Information' table lists the following details:

Field Label	Description	Object Name	Data Type
Description	Description	Project_Task_Template	Long Text Area
API Name	Description__c		
Description			
Help Text			
Data Owner			
Field Usage			
Data Sensitivity Level			
Compliance Categorization			
Created By	AKKAPU SATYA, 27/12/2025, 8:11 am	Modified By	AKKAPU SATYA, 27/12/2025, 8:11 am

The screenshot shows the Salesforce Setup interface for the 'Project Task Template' custom field, specifically the 'Details' tab. The left sidebar is the same as the previous screenshot. The main content area is titled 'Project Task Template Details' and includes 'Edit' and 'Delete' buttons. The 'Details' section lists the following information:

Field Label	Description	Object Name	Data Type
Description	Description	Project_Task_Template	Long Text Area
API Name	Project_Task_Template__c		
Custom			
Singular Label	Project Task Template		
Plural Label	Project Task Templates		
Enable Reports			
Track Activities			
Track Field History			
Deployment Status	Deployed		
Help Settings			
Standard salesforce.com Help Window			

Phase 4 — Process Automation (Admin):

4.1 Overview

Phase 4 focuses on automating business processes in the Project Management CRM using Salesforce declarative tools. Automation reduces manual effort, enforces data accuracy, improves response time, and ensures standardized workflows across the organization. The automation components implemented include Validation Rules, Workflow Rules, Process Builder, Approval Processes, Flow Builder, Email Alerts, Field Updates, Task Automation, and Custom Notifications.

4.2 Validation Rules

Use Case

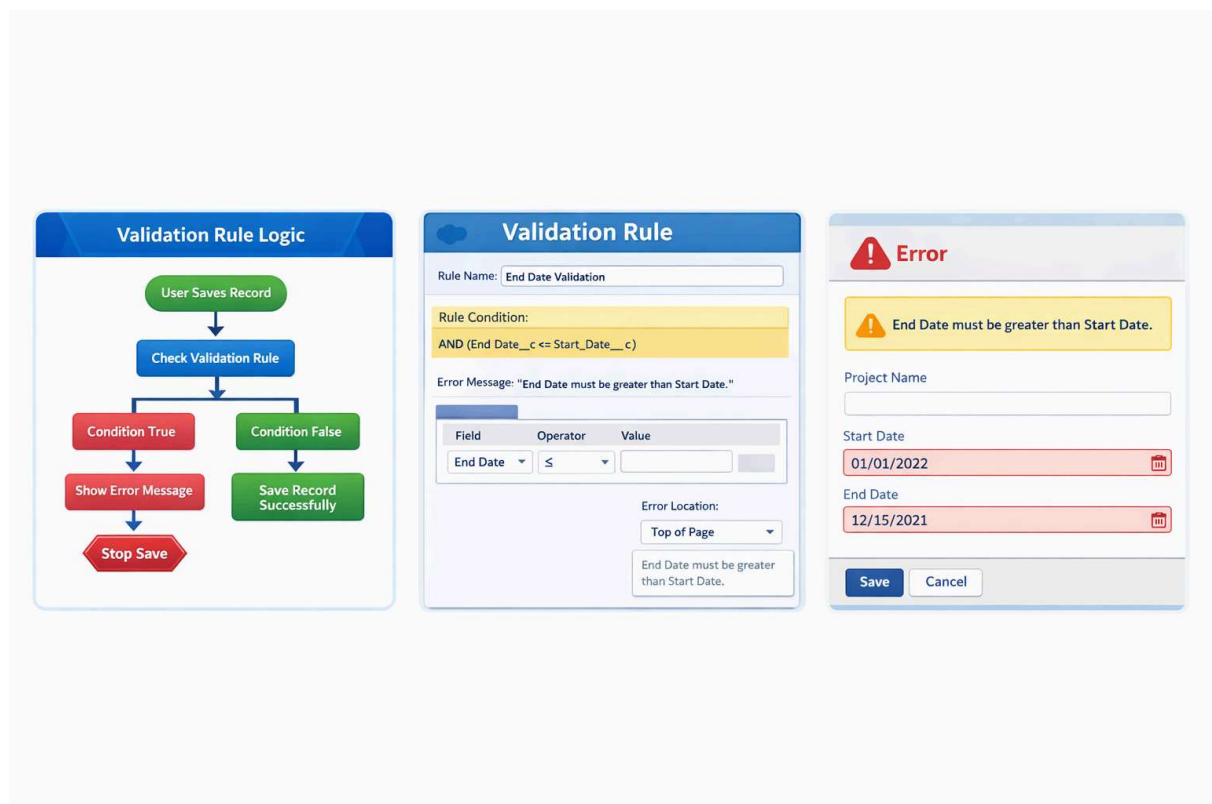
To ensure data integrity, validation rules are implemented on Project and Task objects. For example, users cannot save a Project record without entering Start Date, End Date, and Project Manager. Additionally, End Date must always be greater than Start Date.

Description

Validation rules prevent incorrect or incomplete data from being saved. If a user violates the condition, Salesforce displays a custom error message.

Example Validations

- Start Date cannot be blank
- End Date must be after Start Date
- Task Status cannot be “Completed” if Progress < 100%



4.3 Workflow Rules

Use Case

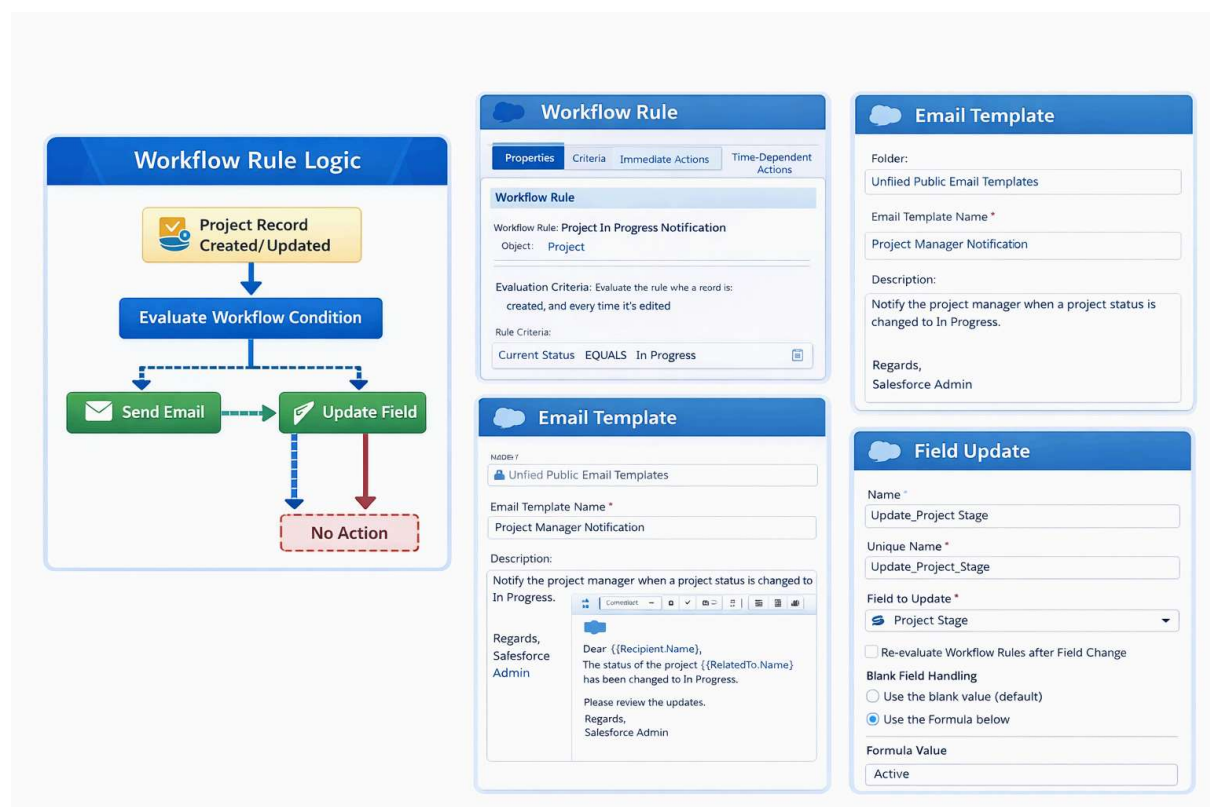
When a Project Status changes to “In Progress”, automatically send an email notification to the Project Manager and update the Project Stage field.

Description

Workflow rules automatically trigger actions based on field updates or record creation.

Automated Actions

- Email Alert to Project Manager
- Field Update → Project Stage = Active



4.4 Process Builder

Use Case

Automatically create a Task for the assigned user when a new Project is created.

Description

Process Builder is used for complex if-then logic with multiple actions.

Actions Performed

- Create Task
- Update Project Status
- Send Notification

4.5 Approval Process

Use Case

When Project Budget exceeds a predefined limit, the record must be approved by a Manager before activation.

Description

The approval process ensures controlled decision-making and authorization.

Steps

- Submit for Approval
- Manager Approves / Rejects
- Status Updated Automatically

4.6 Flow Builder

a) Record-Triggered Flow

Use Case:

When a Task is marked Completed, automatically update Project Progress.

b) Screen Flow**Use Case:**

Guided screen for creating Project and initial Tasks together.

c) Scheduled Flow**Use Case:**

Send reminder emails for overdue tasks daily.

4.7 Email Alerts

Use Case

Email sent when a task is assigned or project status changes.

Description

Uses email templates and workflow/process actions.

4.8 Field Updates

Use Case

Automatically set Project Stage to “Closed” when Status = Completed.

4.9 Tasks Automation

Use Case

Auto-create follow-up task when project reaches 80% completion.

4.10 Custom Notifications

Use Case

Notify team members when a high-priority task is assigned.

Description

Custom notifications improve real-time communication inside Salesforce.

Benefits of Phase 4 Automation

- Reduces manual work
- Improves data accuracy
- Ensures process compliance
- Faster communication
- Better user experience

Phase 5 – Apex Programming (Developer)

5.1 Overview

Phase 5 focuses on implementing custom business logic using Salesforce Apex programming. Apex is used when declarative tools are not sufficient to meet complex business requirements. In the Project Management CRM, Apex is utilized to automate task creation, calculate project progress, enforce business rules, handle asynchronous operations, and integrate scalable processing.

5.2 Apex Classes & Objects

Use Case

An Apex class is created to manage project-related operations such as calculating project progress and validating business rules.

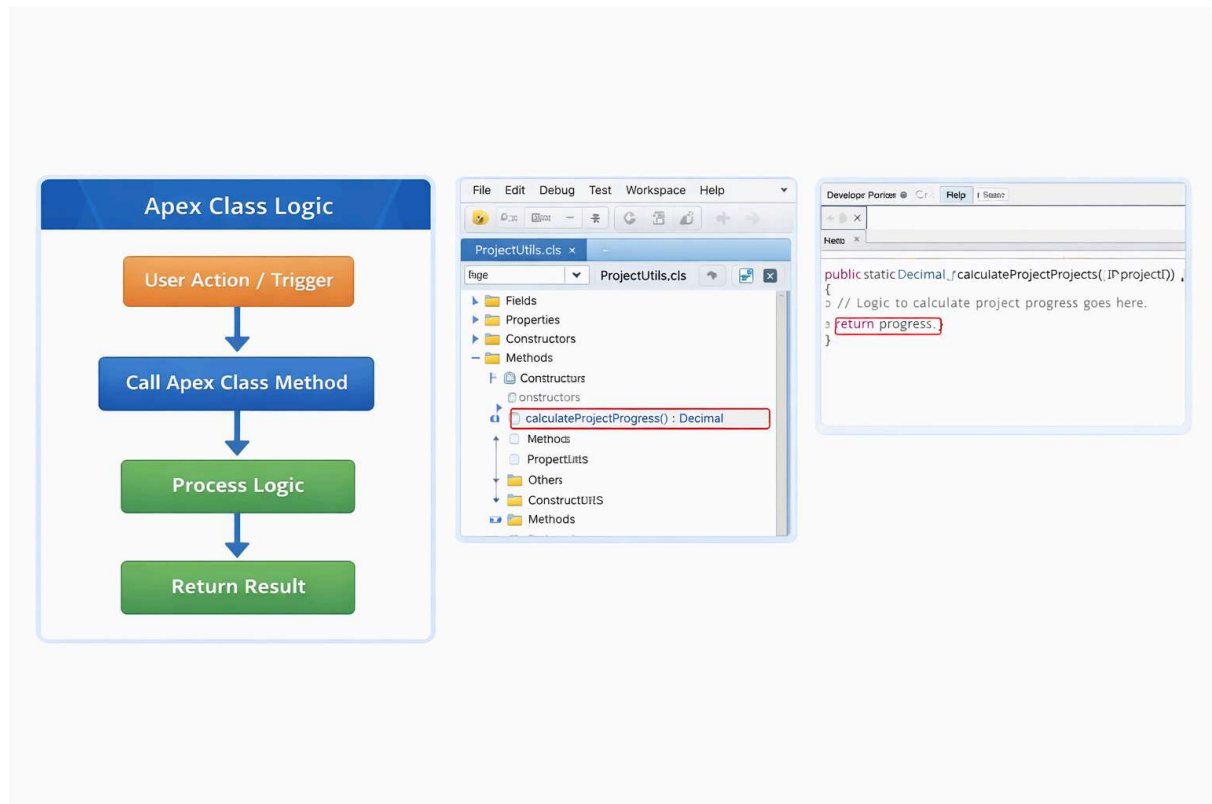
Description

Apex classes encapsulate reusable logic and allow modular development.

Example Functionality

- Calculate project completion percentage

- Validate project dates
- Fetch related tasks



5.3 Apex Triggers (Before & After)

Use Case

When a Task record is inserted or updated, update the Project Progress field automatically.

Description

Triggers execute automatically before or after data operations.

Trigger Events Used

- Before Insert
- After Update

5.4 Trigger Design Pattern

Use Case

Separate trigger logic and business logic using handler classes.

Description

Improves maintainability and scalability.

5.5 SOQL & SOSL

Use Case

Fetch all tasks related to a project and search projects by keyword.

Description

SOQL retrieves records; SOSL searches across objects.

5.6 Collections (List, Set, Map)

Use Case

Store multiple Task IDs in a List and map Project IDs with Tasks.

Description

Collections optimize bulk processing.

5.7 Control Statements

Use Case

Use IF-ELSE to check task status and FOR loop to iterate records.

5.8 Batch Apex

Use Case

Recalculate progress for all projects nightly.

Description

Processes large datasets in chunks.

5.9 Queueable Apex

Use Case

Send bulk notifications asynchronously.

5.10 Scheduled Apex

Use Case

Run automation every day at midnight.

5.11 Future Methods

Use Case

Perform callout to external service without blocking the user.

5.12 Exception Handling

Use Case

Display user-friendly messages if an error occurs.

5.13 Test Classes

Use Case

Ensure minimum 75% code coverage.

5.14 Asynchronous Processing

Use Case

Long-running operations handled in background.

Benefits of Phase 5

- Handles complex logic
- Improves system performance
- Ensures scalability
- Enhances reliability

Phase 6 – User Interface Development

6.1 Overview

Phase 6 focuses on designing and implementing a user-friendly and responsive interface for the Project Management CRM using Salesforce Lightning Experience. The objective of this phase is to provide users with intuitive navigation, personalized pages, interactive components, and seamless access to project and task information. Lightning App Builder and Lightning Web Components (LWC) are used to create modern, dynamic, and efficient user interfaces.

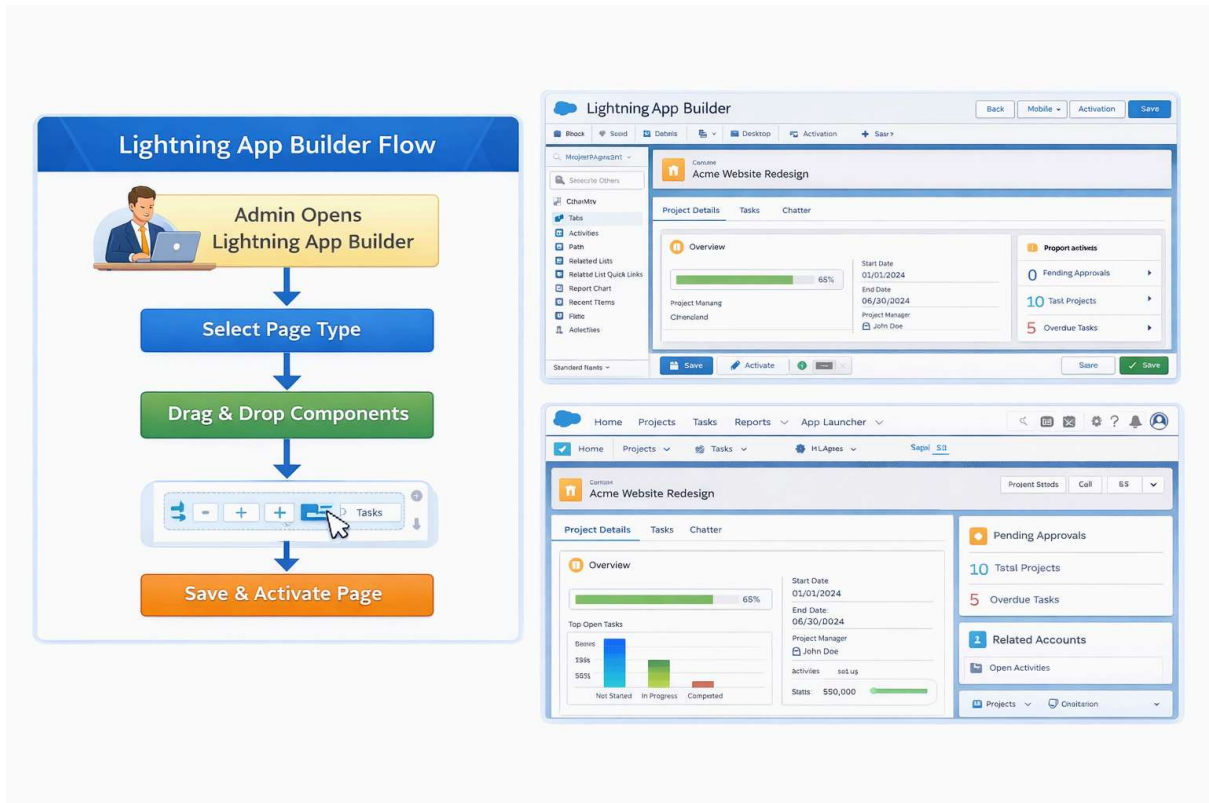
6.2 Lightning App Builder

Use Case

Create a custom Project Management App interface with tailored pages for Projects, Tasks, and Dashboard.

Description

Lightning App Builder allows administrators to design custom pages using drag-and-drop components without coding.



6.3 Record Pages

Use Case

Display Project details, related Tasks, and progress chart on a single page.

Description

Record Pages provide contextual information for each record.

6.4 Tabs

Use Case

Create custom tabs for Projects and Tasks.

6.5 Home Page Layouts

Use Case

Show key metrics such as Total Projects, Overdue Tasks, and Pending Approvals.

6.6 Utility Bar

Use Case

Provide quick access to Notes, Recent Items, and Notifications.

6.7 Lightning Web Components (LWC)

Use Case

Create a custom component to display Project Progress Bar.

Description

LWC provides fast, reusable UI components.

6.8 Apex with LWC

Use Case

Fetch project details from Apex controller.

6.9 Events in LWC

Use Case

Refresh component when task is updated.

6.10 Wire Adapters

Use Case

Retrieve project records automatically.

6.11 Imperative Apex Calls

Use Case

Fetch tasks on button click.

6.12 Navigation Service

Use Case

Navigate from Project list to Task record.

Benefits of Phase 6

- Improved user experience
- Faster navigation
- Modern UI
- Higher productivity

Phase 7 – Integration & External Access

7.1 Overview

Phase 7 focuses on integrating the Project Management CRM with external systems and enabling secure communication between Salesforce and third-party applications. This phase ensures seamless data exchange, secure authentication, and real-time updates using Salesforce integration tools such as Named Credentials, Web Services, Callouts, Platform Events, and Change Data Capture.

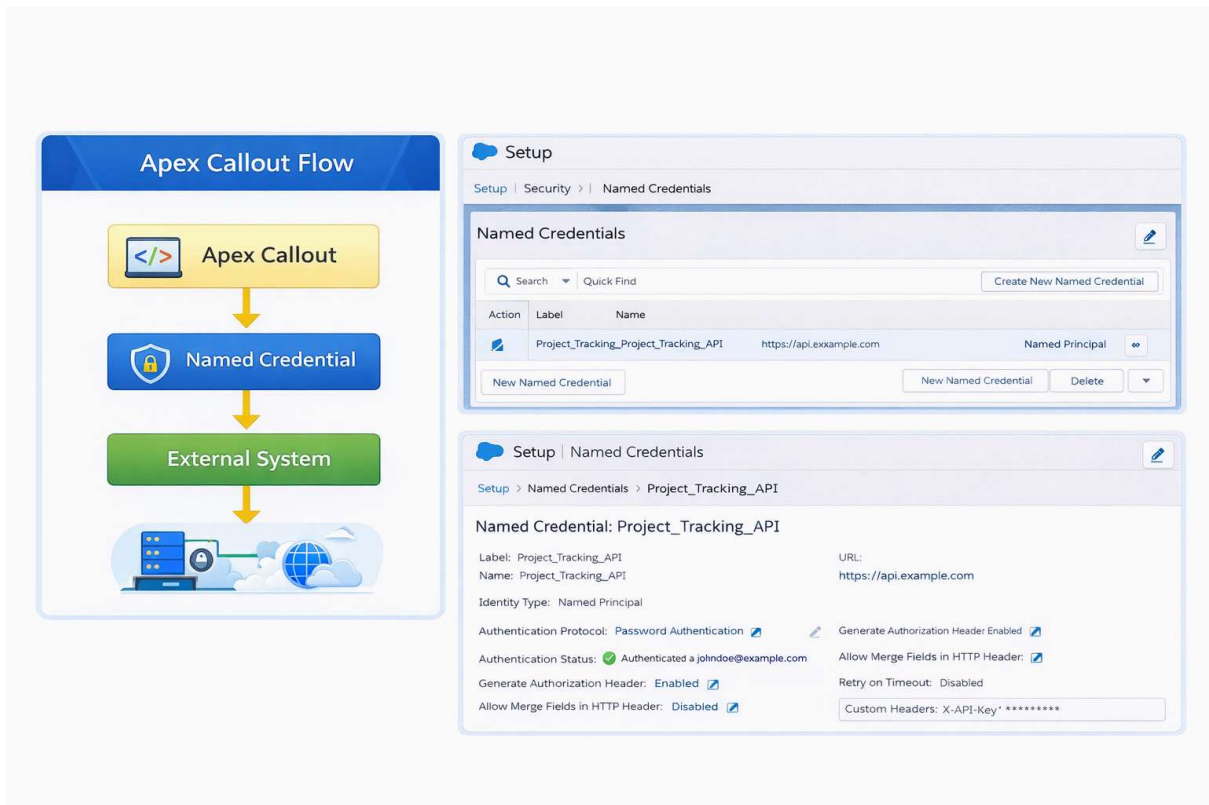
7.2 Named Credentials

Use Case

Store authentication details for an external project tracking API securely.

Description

Named Credentials simplify authentication and endpoint management by securely storing credentials and URLs.



7.3 External Services

Use Case

Automatically generate Apex classes from an external REST API.

Description

External Services allow Salesforce to consume APIs declaratively.

7.4 Web Services (REST / SOAP)

Use Case

Expose project data to an external reporting system.

Description

Salesforce publishes REST APIs to allow external systems to retrieve or update data.

7.5 Callouts

Use Case

Send project status to the external system when status changes.

Flowchart:

Project Updated

|

Apex Callout

|

External API

|

Response Received

7.6 Platform Events

Use Case

Publish events when a project is completed.

Description

Platform events enable event-driven integration.

7.7 Change Data Capture (CDC)

Use Case

Notify external systems when project records change.

7.8 Salesforce Connect

Use Case

View external database tables inside Salesforce.

7.9 API Limits

Use Case

Monitor API usage to prevent exceeding limits.

7.10 OAuth & Authentication

Use Case

Allow third-party apps to access Salesforce securely.

7.11 Remote Site Settings

Use Case

Whitelist external API endpoint.

Benefits of Phase 7

- Secure integrations
- Real-time data exchange
- Scalable architecture
- Improved system connectivity

Phase 8 – Data Management & Deployment

8.1 Overview

Phase 8 focuses on managing project data effectively and deploying the Salesforce application from development to production environments. This phase ensures data accuracy, security, backup, migration, and smooth deployment using Salesforce tools such as Data Import Wizard, Data Loader, Change Sets, and Metadata API.

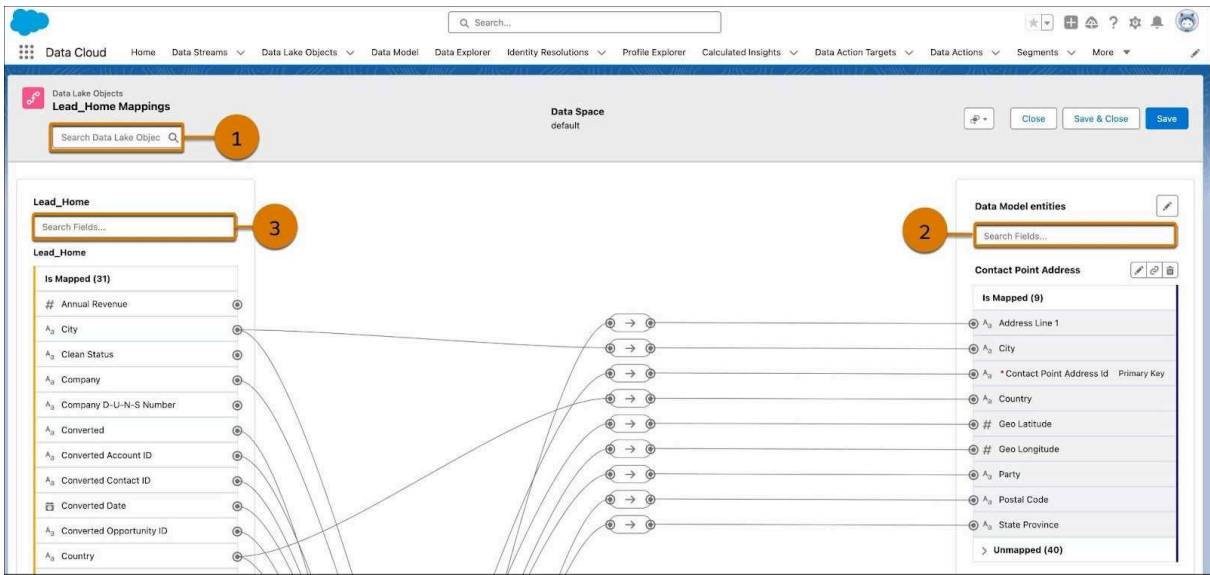
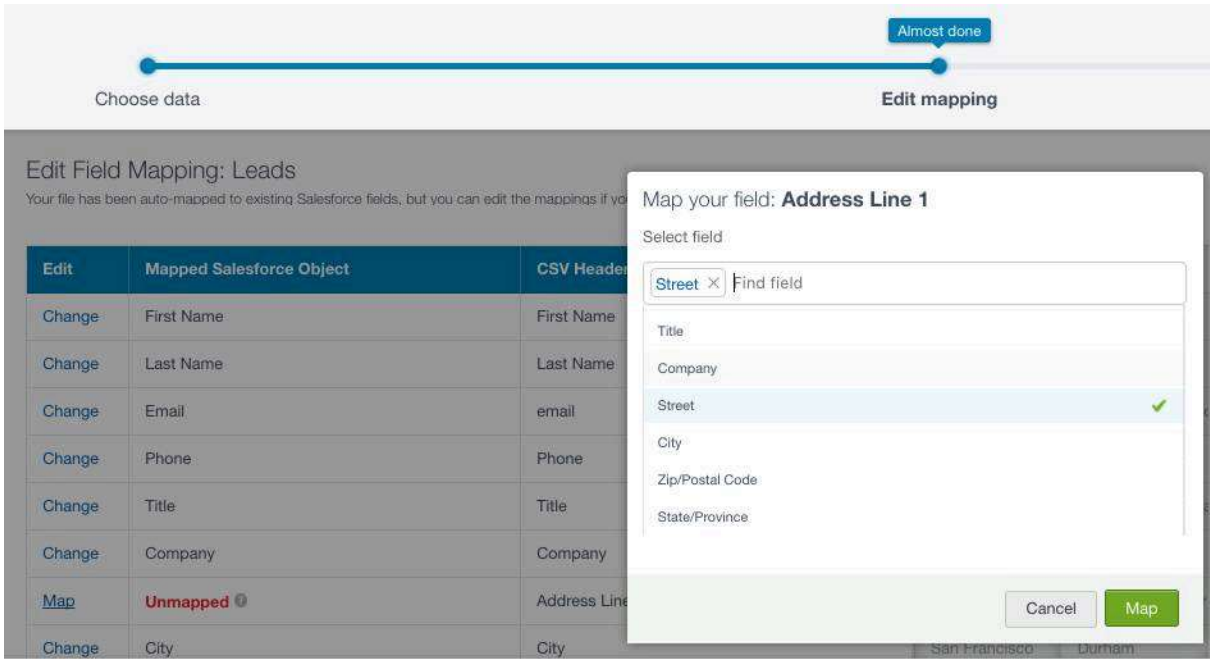
8.2 Data Import Wizard

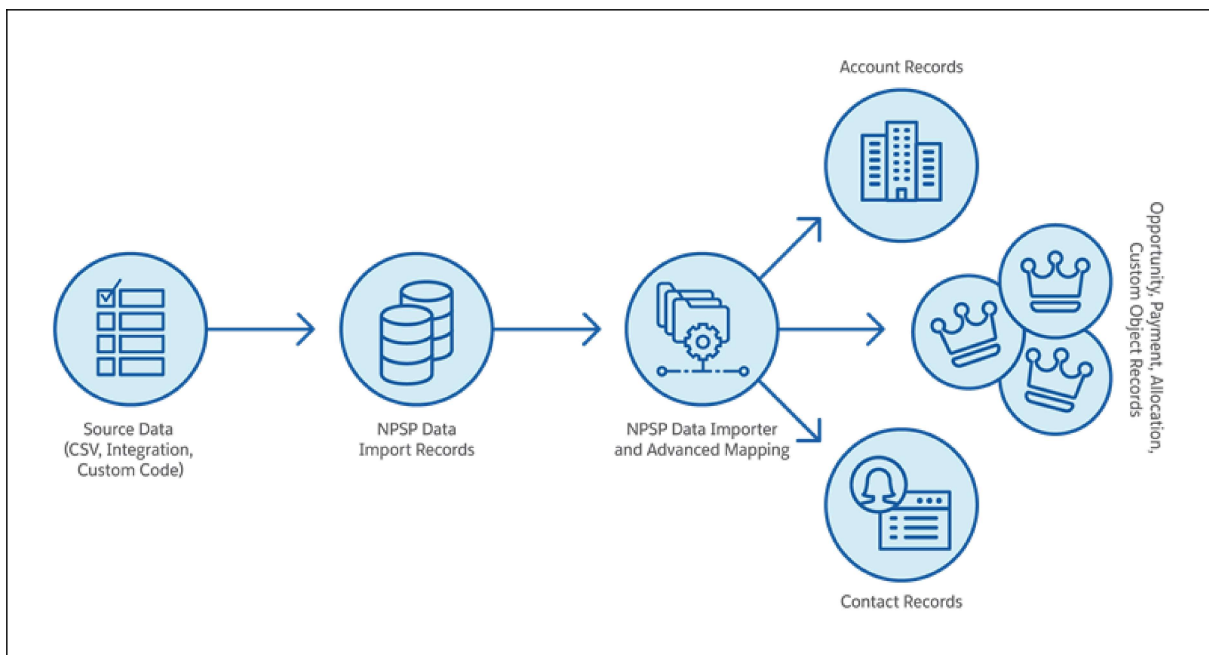
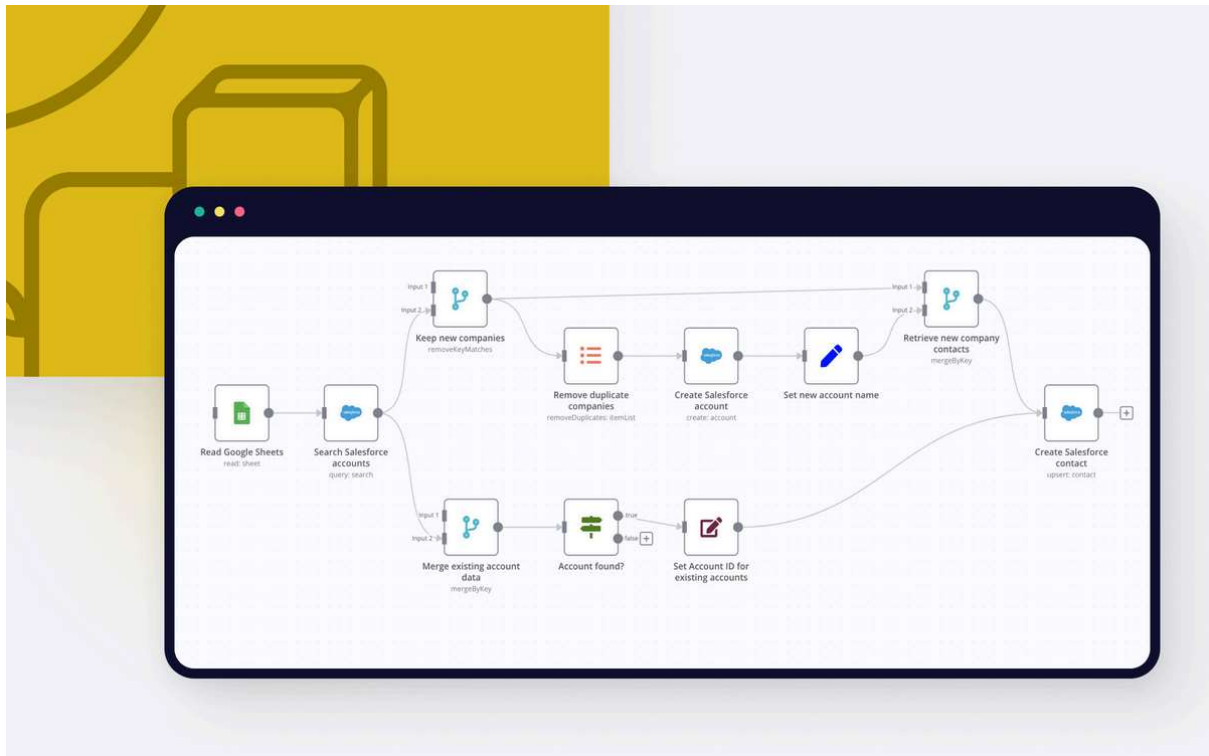
Use Case

Import project records, tasks, and users into Salesforce.

Description

The Data Import Wizard allows administrators to easily upload CSV files and map fields to Salesforce objects.





8.3 Data Loader

Use Case

Insert, update, delete, and export large volumes of project data.

Description

Data Loader is used for handling bulk data operations efficiently.

8.4 Data Backup & Recovery

Use Case

Create backups to prevent data loss.

Description

Regular backups ensure project and task data can be restored in case of system failure.

8.5 Sandbox Management

Use Case

Test new features without affecting production.

Description

Sandboxes are copies of production used for testing and training.

8.6 Change Sets

Use Case

Deploy custom objects, fields, workflows, and Apex code.

8.7 Metadata API Deployment

Use Case

Deploy using VS Code and Salesforce CLI.

8.8 Version Control (Git)

Use Case

Track changes and collaborate.

8.9 Post Deployment Testing

Use Case

Verify that all features work correctly.

8.10 Deployment Checklist

- ✓ Backup completed
- ✓ Sandbox tested
- ✓ Change sets prepared
- ✓ Validation successful
- ✓ Deployment successful

Benefits of Phase 8

- Safe data handling
- Reduced deployment errors
- Easy rollback
- Reliable production releases

Phase 9 – Reporting, Dashboards & Security Review

This phase focuses on transforming Salesforce data into meaningful insights using **Reports & Dashboards** and ensuring data protection using **Security Controls** such as Profiles, Roles, Permission Sets, and Sharing.

Reports & Report Types – Use Case

Reports are used to analyze project performance, task completion status, overdue activities, and resource utilization. Different report formats such as Tabular, Summary, Matrix, and Joined reports allow managers to view data in multiple perspectives for decision-making.

Example Use Case:

A Project Manager views a **Summary Report** showing tasks grouped by project with total hours and completion percentage.

Flowchart

Select Report Type

- Choose Object
- Apply Filters
- Generate Report
- Save & Share

The screenshot displays the Salesforce Reports interface for 'Sunshine Chocolates'. The top navigation bar includes the company logo, a search bar, and various utility icons. The main navigation menu shows 'Sales' as the active section, with sub-menus for Home, Chatter, Projects, Calendar, Opportunities, Accounts, Contacts, Reports, Dashboards, Files, and Tasks.

The 'REPORT' section is titled 'All Opportunities' and 'Opportunities'. It includes a search bar, a 'Filters' button, and a 'Run' button. The report is titled 'Previewing a limited number of records. Run the report to see everything.' and has an 'Update Preview Automatically' toggle.

The report is a Summary Report for 'Opportunities'. It features a 'Fields Pane' on the left with a list of fields including 'Created By', 'Created Alias', 'Last Modified By', 'Last Modified Alias', 'Opportunity Name', 'Type', 'Lead Source', 'Primary Partner', 'Amount', 'Expected Revenue', 'Closed', 'Won', 'Close Date', 'Close Date (2)', 'Close Month', 'Last Stage Change Date', 'Next Step', 'Stage', 'Probability (%)', and 'Fiscal Period'. The 'Outline and Filter Pane' is also visible, showing 'Groups' and 'Columns'.

The 'Report Preview Pane' displays a table with columns: 'Account Name', 'Opportunity Name', 'Amount', 'Expected Revenue', 'Completion Percentage', and 'Close Date'. The table is grouped by 'Account Name' and 'Opportunity Name'. The 'Report Footer' at the bottom shows 'Row Counts', 'Detail Rows', 'Subtotals', and 'Grand Total'.

Account Name	Opportunity Name	Amount	Expected Revenue	Completion Percentage	Close Date
EatWell Provisions (1)	EatWell Provisioning	\$7,500.00		100%	12/1/2023
Subtotal					
Express Airlines (3)	Express Airlines Welcome bags	\$7,500.00	\$3,750.00	50%	12/1/2023
	Express Airlines Dessert Addition	\$12,500.00	\$9,375.00	75%	10/1/2023
	Express Airlines Corporate Retreat Gift Bags	\$22,000.00	\$22,000.00	100%	10/1/2023
Subtotal					
Grand Hotels & Resorts Ltd (2)	Grand Hotels Custom C	\$1,125.00	\$1,125.00	75%	9/1/2023
	Grand Hotels Custom C	\$1,500.00	\$1,500.00	20%	10/1/2023
Subtotal					
Luxury Hospitality (1)	Luxury Hospitality	\$5,000.00	\$5,000.00	100%	12/1/2023
Subtotal					
Mary's Gourmet Foods (2)	Mary's Gourmet Foods - 2nd Store	\$22,000.00	\$4,400.00	20%	12/1/2023
	Mary's Gourmet Foods Intro package	\$18,000.00	\$18,000.00	100%	10/1/2023
Subtotal					
United Foods Corp (8)	United Foods	\$17,000.00	\$1,700.00	10%	10/1/2023
	United Foods	\$17,000.00	\$17,000.00	100%	10/1/2023
	United Foods	\$17,000.00	\$17,000.00	100%	10/1/2023
	United Foods	\$22,000.00	\$19,800.00	90%	12/1/2023
Subtotal					

All

Search...

Sales

Home

Opportunities

Leads

Tasks

Files

Accounts

Contacts

Campaigns

Dashboards

Reports

Chatter

Groups

More

REPORT

New Opportunities Report

Opportunities

Got Feedback?

↩

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Add Chart

Save & Run

Save

Close

Run

Outline

Filters

Previewing a limited number of records. Run the report to see everything.

Update Preview Automatically

Opportunity Owner	Account Name	Opportunity Name	Stage	Fiscal Period	Amount	Expected Revenue
1 Emily Call	Dickenson plc	Dickenson Mobile Generators	Qualification	Q3-2007	\$15,000.00	\$1,500.00
2 Emily Call	United Oil & Gas Corp.	United Oil Office Portable Generators	Negotiation/Review	Q3-2007	\$125,000.00	\$112,500.00
3 Emily Call	Express Logistics and Transport	Express Logistics Standby Generator	Closed Won	Q3-2007	\$220,000.00	\$220,000.00
4 Emily Call	GenePoint	GenePoint Standby Generator	Closed Won	Q3-2007	\$85,000.00	\$85,000.00
5 Emily Call	Grand Hotels & Resorts Ltd	Grand Hotels Kitchen Generator	Id. Decision Makers	Q3-2007	\$15,000.00	\$9,000.00
6 Emily Call	United Oil & Gas Corp.	United Oil Refinery Generators	Proposal/Price Quote	Q3-2007	\$270,000.00	\$202,500.00
7 Emily Call	United Oil & Gas Corp.	United Oil SLA	Closed Won	Q3-2007	\$120,000.00	\$120,000.00
8 Emily Call	Grand Hotels & Resorts Ltd	Grand Hotels Guest Portable Generators	Value Proposition	Q3-2007	\$250,000.00	\$125,000.00
9 Emily Call	Edge Communications	Edge Emergency Generator	Closed Won	Q3-2007	\$75,000.00	\$75,000.00
10 Emily Call	University of Arizona	University of AZ Portable Generators	Closed Won	Q3-2007	\$50,000.00	\$50,000.00
11 Emily Call	Pyramid Construction Inc.	Pyramid Emergency Generators	Prospecting	Q3-2007	\$100,000.00	\$10,000.00
12 Emily Call	Express Logistics and Transport	Express Logistics Portable Truck Generators	Value Proposition	Q3-2007	\$80,000.00	\$40,000.00
13 Emily Call	GenePoint	GenePoint Lab Generators	Id. Decision Makers	Q3-2007	\$60,000.00	\$36,000.00
14 Emily Call	GenePoint	GenePoint SLA	Closed Won	Q3-2007	\$30,000.00	\$30,000.00
15 Emily Call	United Oil & Gas Corp.	United Oil Installations	Negotiation/Review	Q3-2007	\$270,000.00	\$243,000.00

Report: Opportunities

Opportunities by Sum of Amount

Enable Field Editing

Search

Add Chart

Filter

Refresh

Reset

Edit

Close Month	Type	New Business	Existing Business	New Business / Add-on	Total
2/1/2023	Sum of Amount Record Count	\$11,770,199.00 7	\$10,325,088.00 5	\$7,130,027.00 9	\$29,225,314.00 21
3/1/2023	Sum of Amount Record Count	\$13,873,765.00 10	\$4,923,100.00 2	\$3,148,166.00 6	\$21,945,031.00 18
4/1/2023	Sum of Amount Record Count	\$7,422,779.00 11	\$287,850.00 1	\$3,358,564.00 3	\$11,069,193.00 15
5/1/2023	Sum of Amount Record Count	\$13,706,979.00 13	\$3,314,900.00 1	\$1,420,523.00 4	\$18,442,402.00 18
6/1/2023	Sum of Amount Record Count	\$10,238,814.00 11	\$6,929,477.00 9	\$9,932,512.00 8	\$27,100,803.00 28
7/1/2023	Sum of Amount Record Count	\$26,803,617.00 19	\$8,395,230.00 9	\$15,307,946.00 14	\$50,506,793.00 42
8/1/2023	Sum of Amount Record Count	\$11,703,987.00 15	\$8,272,525.00 5	\$13,507,666.00 14	\$33,484,178.00 34
9/1/2023	Sum of Amount Record Count	\$20,369,912.00 17	\$6,878,485.00 5	\$10,530,468.00 9	\$37,778,865.00 31
10/1/2023	Sum of Amount Record Count	\$14,240,108.00 15	\$13,889,590.00 8	\$13,531,634.00 12	\$41,661,332.00 35
11/1/2023	Sum of Amount Record Count	\$7,144,844.00 12	\$7,872,940.00 8	\$18,631,767.00 13	\$33,649,551.00 33
12/1/2023	Sum of Amount Record Count	\$21,369,842.00 22	\$5,869,320.00 7	\$7,504,939.00 16	\$34,744,101.00 45
1/1/2024	Sum of Amount Record Count	\$16,478,314.00 12	\$2,207,200.00 1	\$8,871,362.00 8	\$27,556,876.00 21
Total	Sum of Amount Record Count	\$210,956,782.00 191	\$85,828,945.00 70	\$114,781,144.00 125	\$411,566,871.00 386

REPORT ▼
New Opportunities Report / Opportunities

After

Previewing a limited number of records. Run the report to see everything.

Update Preview Automatically

Stage	Close Date	06/04/2023	07/05/2023	10/05/2023	13/05/2023	16/05/2023	21/05/2023	23/05/2023	29/05/2023	29/05/2023	Total
Proposal/Price Quote	Sum of Amount	0	£0.00	£0.00	£0.00	£2,70,000.00	£0.00	£0.00	£0.00	£0.00	£2,70,000.00
	Record Count	0	0	0	0	1	0	0	0	0	1
Negotiation/Review	Sum of Amount	0	£0.00	£0.00	£0.00	£0.00	£0.00	£0.00	£0.00	£0.00	£1,95,000.00
	Record Count	0	0	0	0	0	0	0	0	0	2
Closed Won	Sum of Amount	0	£2,35,000.00	£8,15,000.00	£2,10,000.00	£3,50,000.00	£0.00	£1,20,000.00	£1,20,000.00	£30,000.00	£34,45,000.00
	Record Count	1	1	1	1	1	0	1	1	1	10
Total	Sum of Amount	0	£2,35,000.00	£8,15,000.00	£2,10,000.00	£3,50,000.00	£2,70,000.00	£1,20,000.00	£1,20,000.00	£30,000.00	£42,10,000.00
	Record Count	1	1	1	1	1	1	1	1	1	16

Details (19 Rows) Click an intersection in the table above to filter details.

Account Name	Type	Owner Role	Opportunity Owner	Fiscal Period	Amount	Expected Revenue	Probable
15 Grand Hotels & Resorts Ltd	Existing Customer - Upgrade	CEO	F Demo	Q2-2015	£1,50,000.00	£1,50,000.00	
16 United Oil & Gas Corp.	Existing Customer - Upgrade	CEO	F Demo	Q2-2015	£1,20,000.00	£1,20,000.00	
17 United Oil & Gas Corp.	Existing Customer - Upgrade	CEO	F Demo	Q2-2015	£1,20,000.00	£1,20,000.00	
18 Genepart	Existing Customer - Upgrade	CEO	F Demo	Q2-2015	£30,000.00	£30,000.00	
19 Edge Communications	New Customer	CEO	F Demo	Q2-2015	£75,000.00	£75,000.00	
20					£42,10,000.00		

Row Counts Detail Rows Grand Total Stacked Summaries Conditional Formatting

Dashboards & Dynamic Dashboards – Use Case

Dashboards provide a visual summary of key metrics such as total projects, completed tasks, overdue tasks, and team workload. Dynamic dashboards allow users to see data based on their own access level.

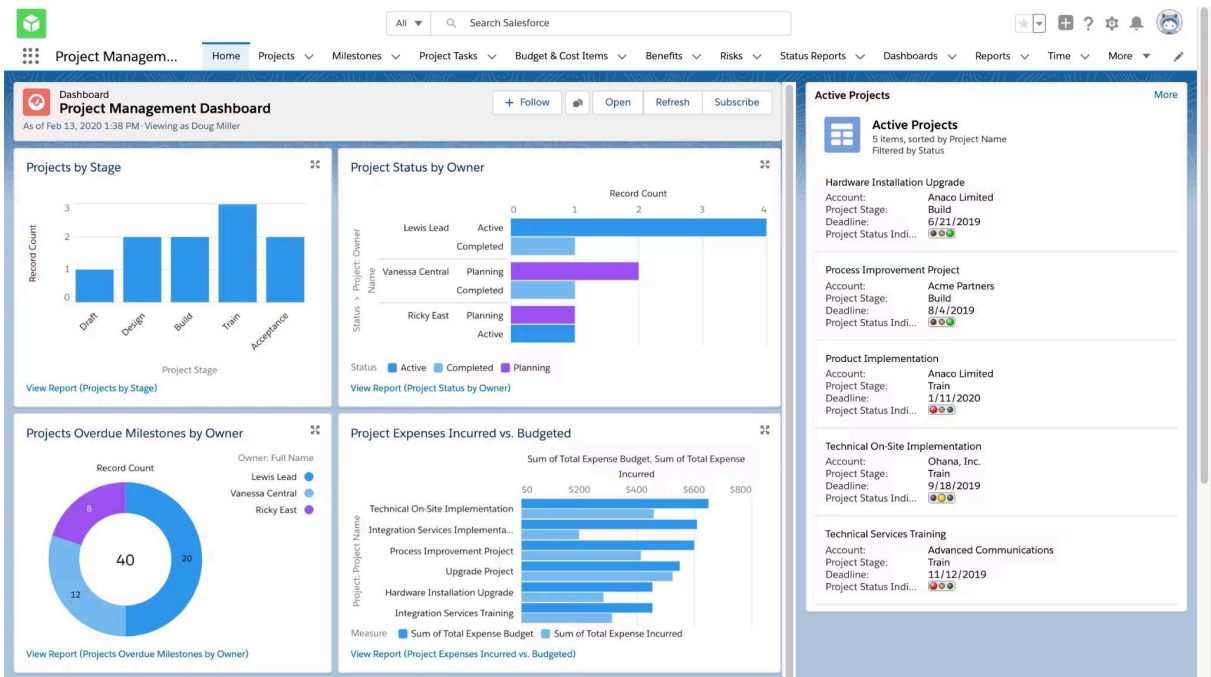
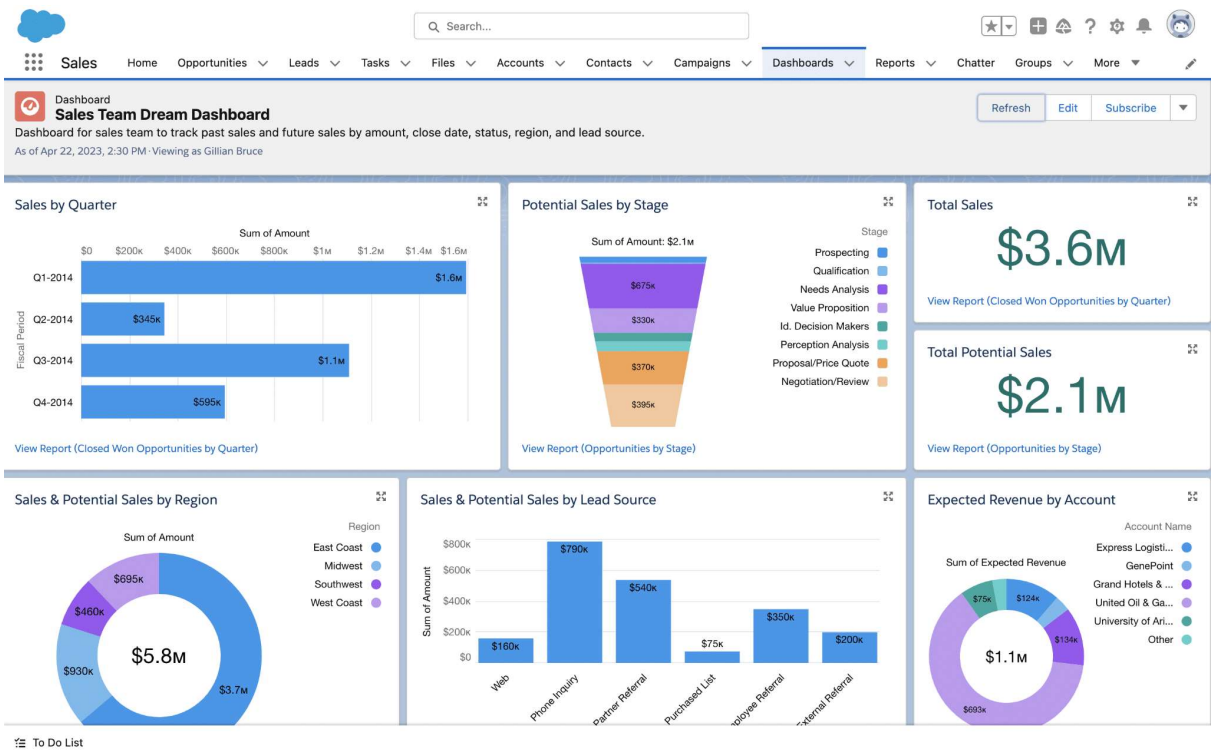
Example Use Case:

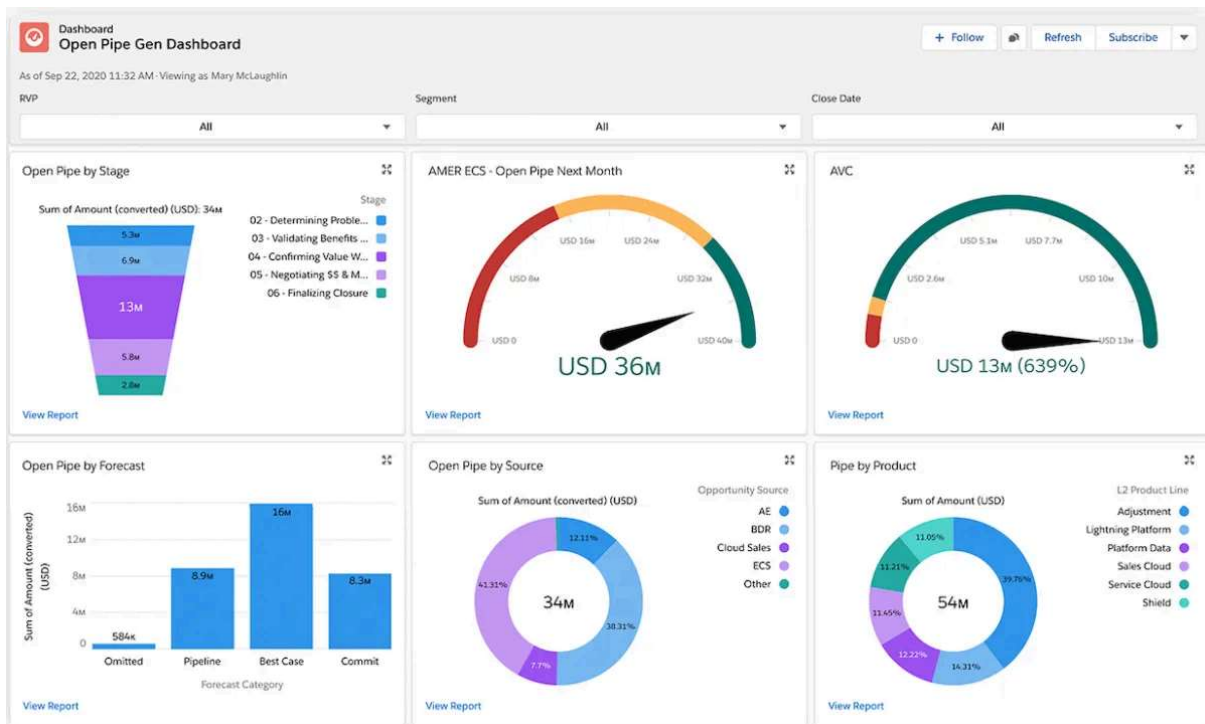
A Team Lead opens a dashboard that displays only their team's project performance.

Flowchart

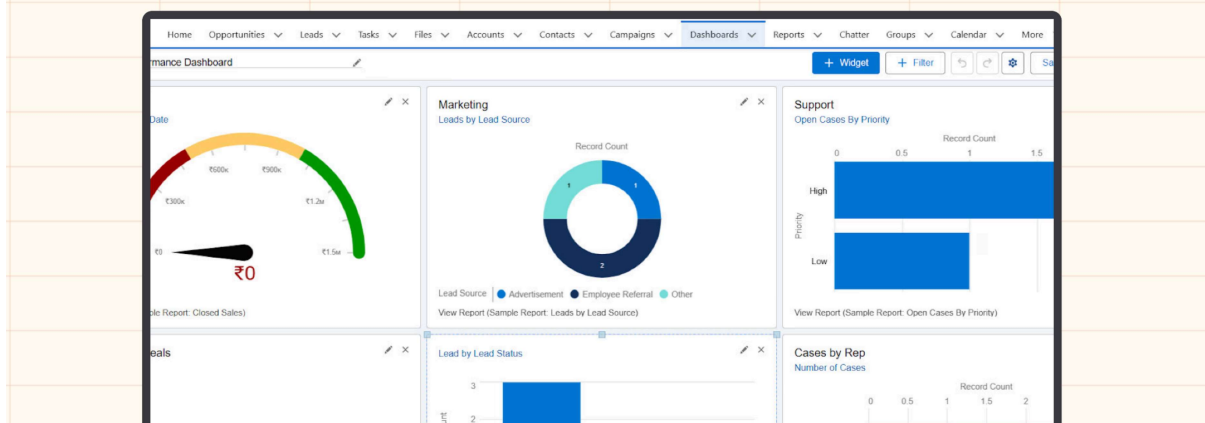
Create Reports

- Create Dashboard
- Add Components
- Choose Data Source
- Save & View





How to Create a Dynamic Dashboard on Salesforce



Profiles, Roles & Permission Sets – Use Case

Security is implemented by defining what each user can view or edit in the system. Profiles provide baseline access, while permission sets grant additional rights.

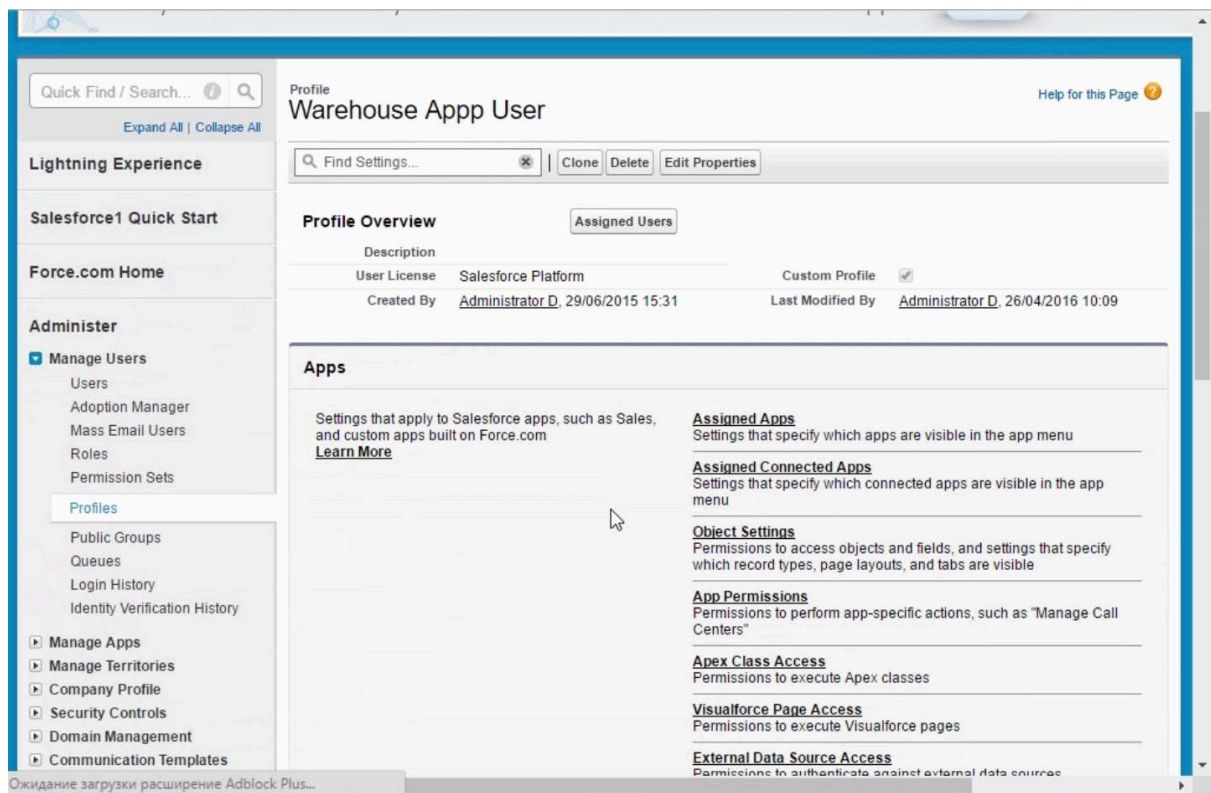
Example Use Case:

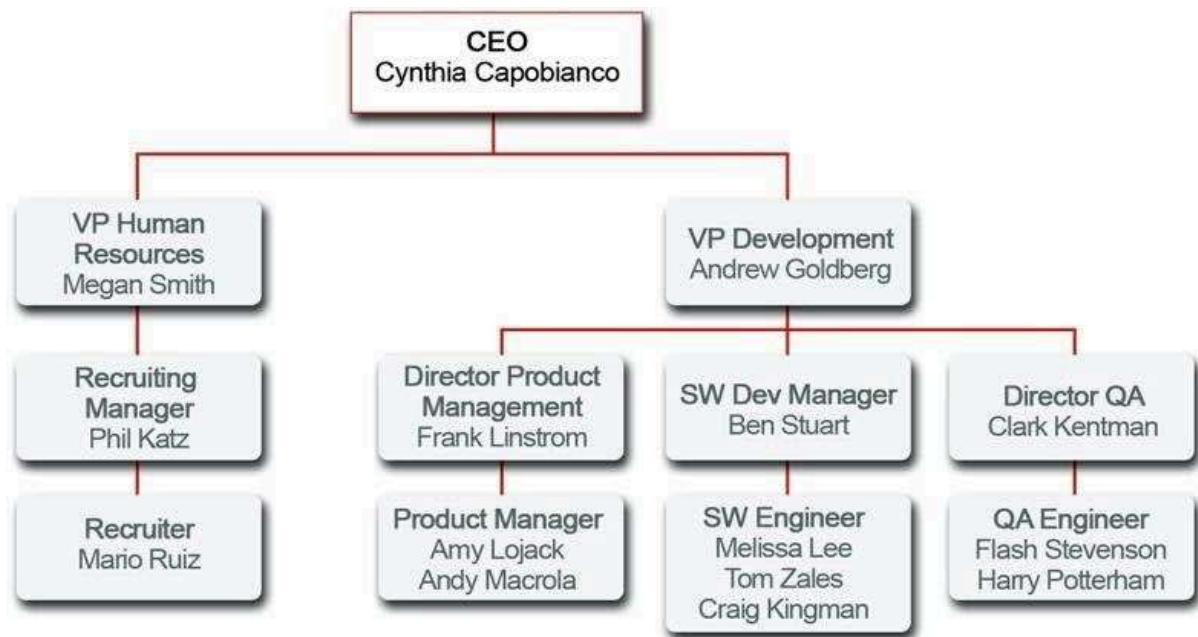
Only Admin can delete projects, while Users can only view and update tasks.

Flowchart

Create Profile

- Assign Permissions
- Create Permission Set
- Assign to User





OWD, Sharing Rules & Field Level Security – Use Case

These features ensure that sensitive project data is visible only to authorized users.

Example Use Case:

The project budget field is visible only to Project Managers and Admins.

Flowchart

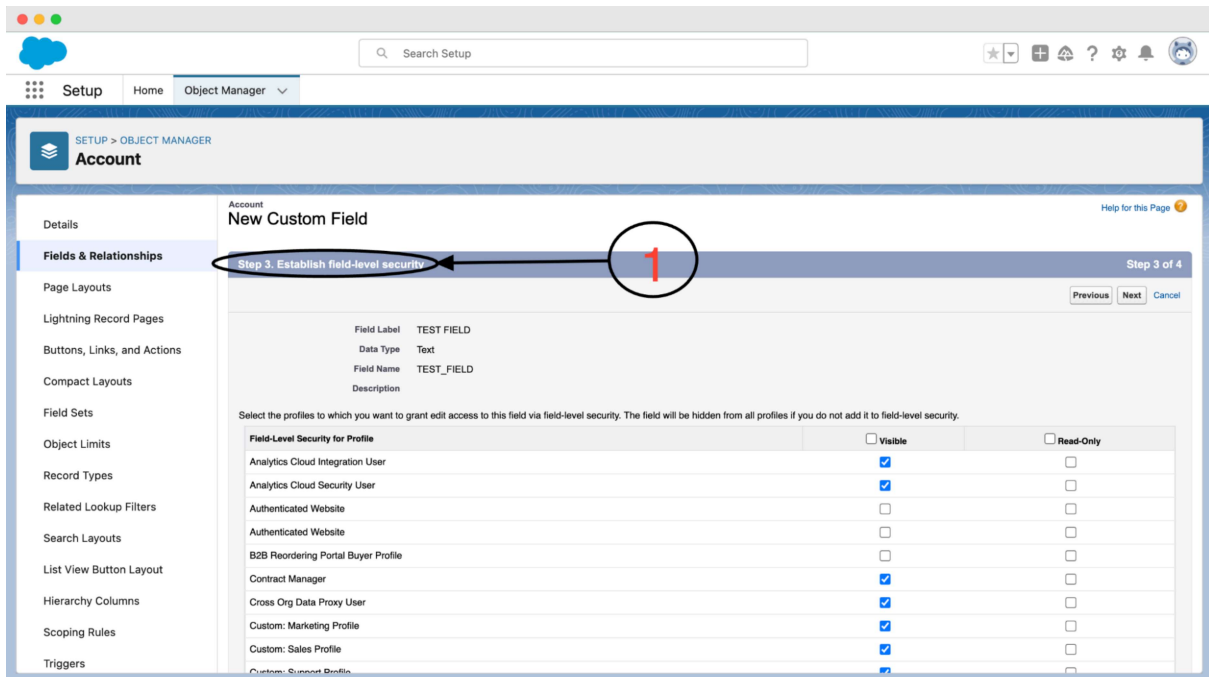
Define OWD

- Create Sharing Rule
- Set Field Level Security
- Test Access

[Organization-Wide Defaults Help](#)

Activate Windows ✓
Go to Settings to activate Windows. ✓

The screenshot shows the Salesforce Setup page for Sharing Settings. The page is titled "Sharing Settings" and includes a sidebar with navigation options like "Setup", "Home", and "Object Manager". The main content area shows the "Sharing Settings" configuration steps. A red circle highlights the "Rule Name" field in Step 1. A red circle highlights the "Based on record owner" radio button in Step 2. A red circle highlights the "Lead: owned by members of" dropdown in Step 3. A red circle highlights the "Share with" dropdown in Step 4. A red circle highlights the "Lead Access" dropdown in Step 5. The page also includes a search bar at the top and a sidebar with a search bar and a list of sharing rules.



Field Permissions

Field Name	Field API Name	☐ Read Access	☐ Edit Access
Apex	Apex__c	☒	☒
C#	C__c	☒	☒
Candidate Name	Name	☒	☒
Created By	CreatedById	☒	☐
Description	Description__c	☒	☒
Email	Email__c	☐	☐
Hire By	Hire_By__c	☒	☐
Hiring Manager	Hiring_Manager__c	☒	☐
Job Category	Job_Category__c	☒	☒
Last Modified By	LastModifiedById	☒	☐
Owner	OwnerId	☒	☒
Phone	Phone__c	☐	☐
Social Security Number	Social_Security_Number__c	☐	☐

Audit Trail & Session Settings – Use Case

Audit Trail tracks configuration changes, while session settings control login behavior and security policies.

Example Use Case:

Admin reviews who modified workflows or objects in the last 24 hours.

Flowchart

User Performs Action

→ System Logs Change

→ Admin Reviews Audit Trail

View Setup Audit Trail		
SETUP View Setup Audit Trail		
6/15/2021, 2:59:47 AM PDT	roopa@astreal.com	Created business process "test"
6/14/2021, 9:44:31 PM PDT	salesforce.com, inc.	Max number of streaming topics
6/14/2021, 6:28:09 AM PDT	roopa@astreal.com	Deleted business process "Sales Stage"
6/14/2021, 6:28:04 AM PDT	roopa@astreal.com	Deleted business process "Donation"
6/14/2021, 6:27:15 AM PDT	roopa@astreal.com	Created business process "Donation"
6/14/2021, 6:26:41 AM PDT	roopa@astreal.com	Created business process "Opportunity"
6/14/2021, 6:24:07 AM PDT	roopa@astreal.com	Created business process "Sales Stage"
6/12/2021, 8:51:06 PM PDT	salesforce.com, inc.	Organization setup action: checkCEUserPermOffOn has changed.
6/12/2021, 8:51:00 PM PDT	salesforce.com, inc.	Organization setup action: rqEmailChangeConfirmPrefOffOn has changed.
6/12/2021, 8:51:00 PM PDT	salesforce.com, inc.	Organization setup action: subscriptionAppEnrolledOffOn has changed.
6/4/2021, 3:06:36 AM PDT	salesforce.com, inc.	Organization setup action: inAppLearningOffOn has changed.
5/27/2021, 1:21:08 AM PDT	roopa@astreal.com	Max number of streaming topics
5/27/2021, 1:06:46 AM PDT	roopa@astreal.com	Deleted Web-to-Lead Auto-Response Rule Test
5/27/2021, 1:05:21 AM PDT	roopa@astreal.com	Customize Leads
5/27/2021, 12:55:27 AM PDT	roopa@astreal.com	Created Web-to-Lead Auto-Response Rule Test
5/27/2021, 12:55:27 AM PDT	roopa@astreal.com	Customize Leads
5/27/2021, 12:55:27 AM PDT	roopa@astreal.com	Deleted Web-to-Lead Auto-Response Rule Test
5/27/2021, 12:55:27 AM PDT	roopa@astreal.com	Customize Leads
5/27/2021, 12:55:27 AM PDT	roopa@astreal.com	Created Web-to-Lead Auto-Response Rule Test
5/27/2021, 12:55:27 AM PDT	roopa@astreal.com	Customize Leads

Setup	Home	Object Manager
profile		
Users		
Profiles		
Didn't find what you're looking for? Try using Global Search.		
Session Settings		
Session Times Out After	24 hours of inactivity 12 hours of inactivity 8 hours of inactivity 4 hours of inactivity 2 hours of inactivity 1 hour of inactivity 30 minutes of inactivity 15 minutes of inactivity	Session Security Level Required at Login
Enable different login policies for community and Salesforce login authentication for internal users	☐	--None--
Relax login IP restrictions	☐	
Skip device activation	☐	
Allow OAuth for Internal Users	☐	
Password Policies		
User passwords expire in	90 days	
Enforce password history	3 passwords remembered	
Minimum password length	8	
Password complexity requirement	Must mix alpha and numeric characters	
Password question requirement	Cannot contain password	
Maximum invalid login attempts	10	
Lockout effective period	15 minutes	
Obscure secret answer for password resets	☐	
Require a minimum 1 day password lifetime	☐	
Don't immediately expire links in forgot password emails	☐	



Admin

Users Login History				All Status ▾
		Rodrigo Reboucas	07/17/15 13:02:33	
		Rodrigo Reboucas	07/14/15 05:54:59	
		Rodrigo Reboucas	07/14/15 05:00:08	
		Rodrigo Reboucas	07/14/15 04:57:35	
		Mary Jane	07/14/15 04:56:35	
		Rodrigo Reboucas	07/14/15 04:54:12	
		John Martin	07/14/15 04:53:57	Restricted IP
		Rodrigo Reboucas	07/14/15 04:51:01	

Acme Co | Salesforce

site-velocity-84332.lightning.force.com/lightning/r/Account/0013F00000hZk8mQAC/view

Search...

Sales Home Opportunities Leads Tasks Files Accounts Contacts Campaigns Dashboards Reports Chatter Groups Calendar People Cases More

Account Acme Co

+ Follow Assign AM Assign CSM Delete

Type Prospect Phone Website <https://acme.com> Account Owner [Chloe Closer](#) Industry Billing Address

Related Details News

We found no potential duplicates of this Account.

Account History (3)

Date	Field	User	Original Value	New Value
4/27/2021, 8:10 PM	Account Owner	User User	Sarah Sales	Chloe Closer
4/27/2021, 8:10 PM	Account Owner	User User	User User	Sarah Sales
4/27/2021, 7:41 PM	Created	User User		

[View All](#)

Contacts (0)

New

Opportunities (0)

New

Cases (0)

New

Partners (0)

New

Notes & Attachments (0)

Upload Files

Activity Chatter

New Task New Event Log a Call Email

Create new...

Add

Filters: All time • All activities • All types

Refresh • Expand All • View All

Upcoming & Overdue

No next steps.

To get things moving, add a task or set up a meeting.

No past activity. Past meetings and tasks marked as done show up here.

Phase 10 – Quality Assurance Testing

Quality Assurance (QA) testing ensures that all Salesforce features implemented in the Project Management CRM work correctly, reliably, and according to business requirements. Testing was performed on functional, automation, security, and integration features using both positive and negative scenarios.

Testing Approach

- Manual functional testing for UI components
- Data validation testing
- Automation testing (Flows, Triggers, Workflows)
- Security testing (Profiles, Roles, Permission Sets)
- Integration testing (Callouts, Named Credentials)
- Regression testing after changes

Test Case 1 – Create Project Record

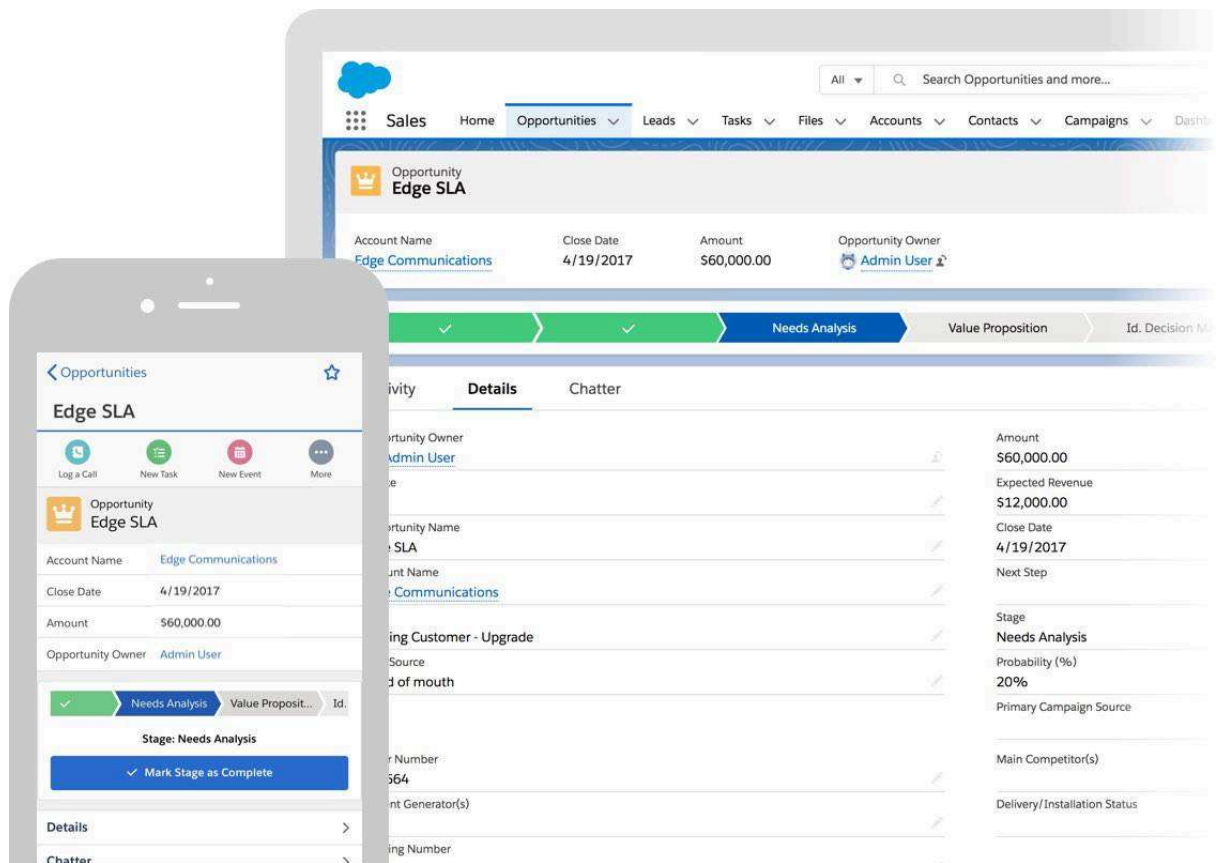
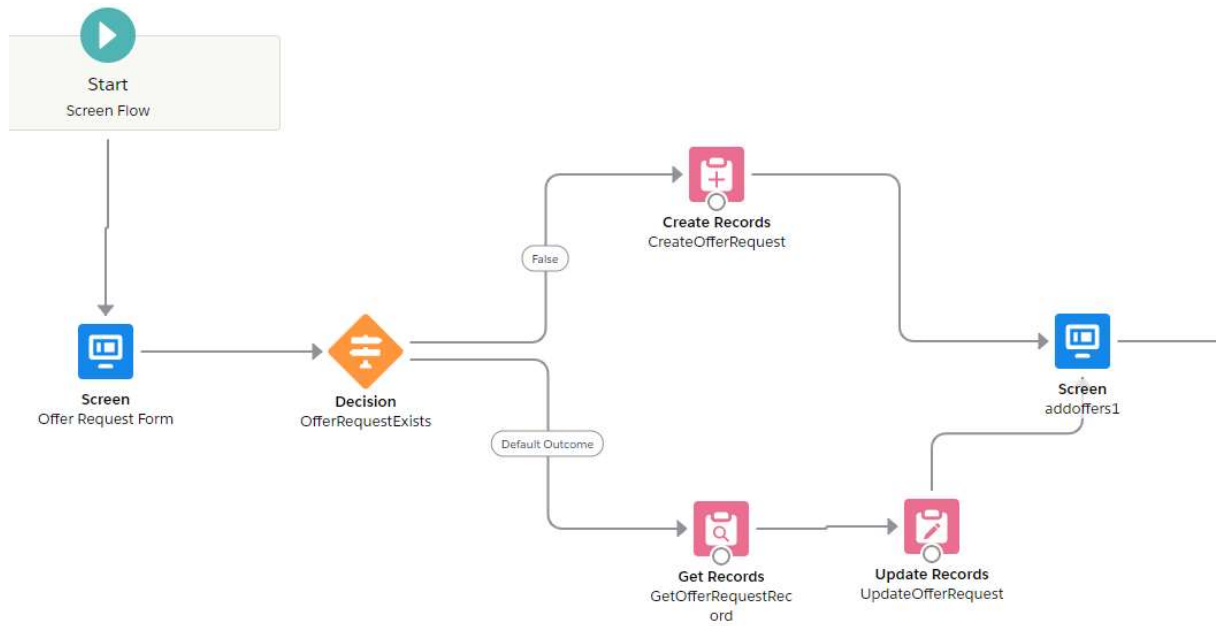
Use Case

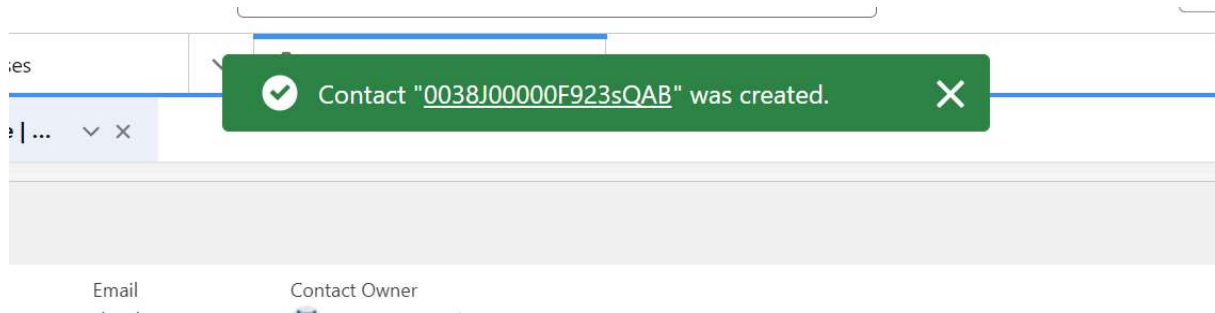
Verify that a user can successfully create a new Project record.

Flowchart

User Enters Project Details

- Click Save
- System Validates Data
- Project Created





Enter Action Information Save Cancel

Object Name i

Action Type i

Target Object i

Standard Label Type i

Label

Name i

Description i

Create Feed Item ☒ i

Success Message i

Icon [Change Icon](#)

Parameter	Description
Input	Project Name = Website App, Status = New
Expected Result	Project record saved
Actual Result	Project record saved successfully
Screenshot	Project Created Screen

Test Case 2 – Validation Rule Testing

Use Case

Verify error message appears when mandatory fields are blank.

Flowchart

Leave Field Blank

- Click Save
- Validation Rule Fires
- Error Message

Test Case 3 – Flow Automation Testing

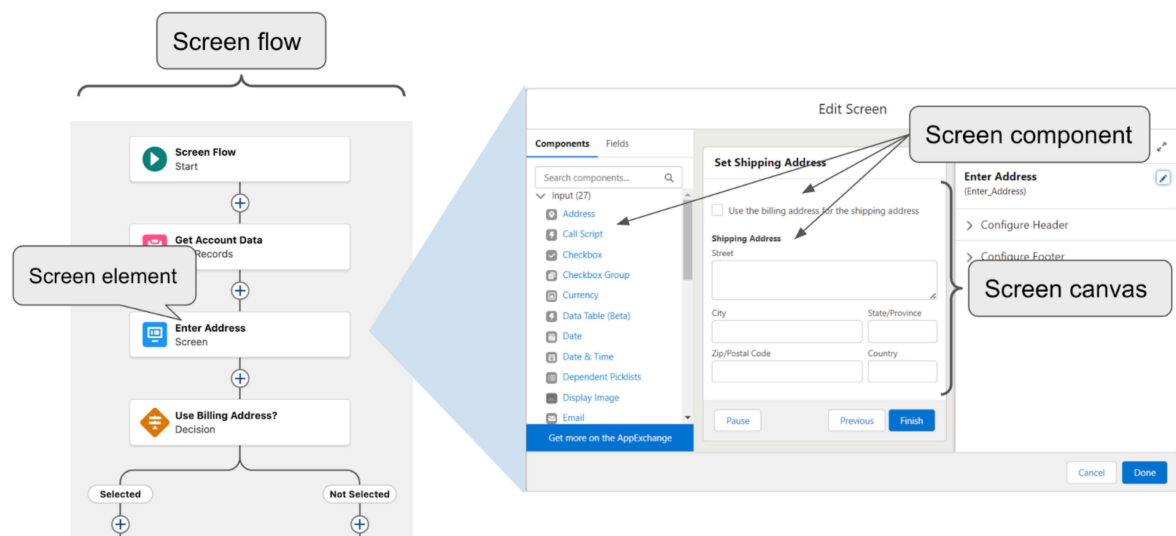
Use Case

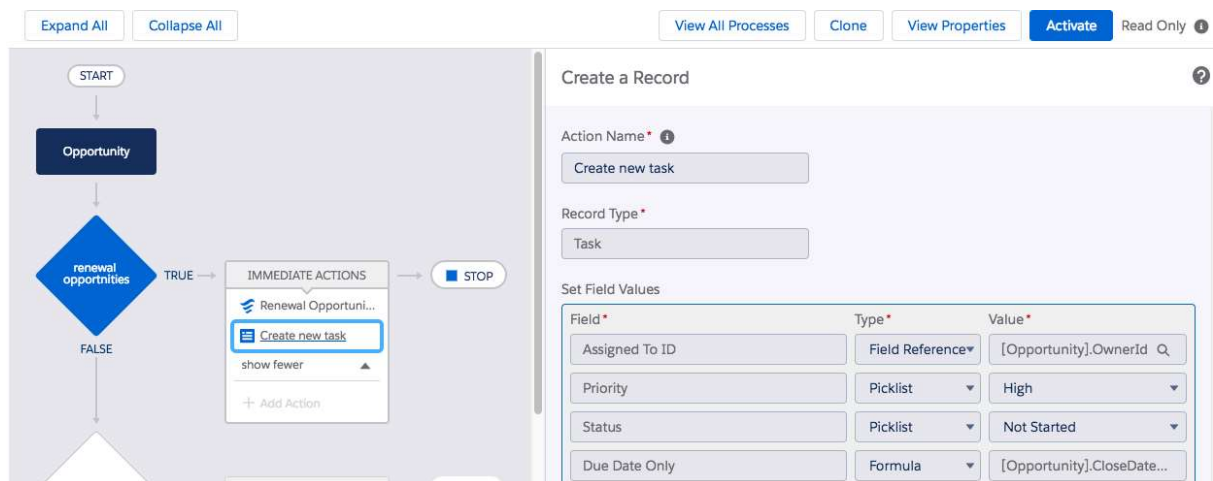
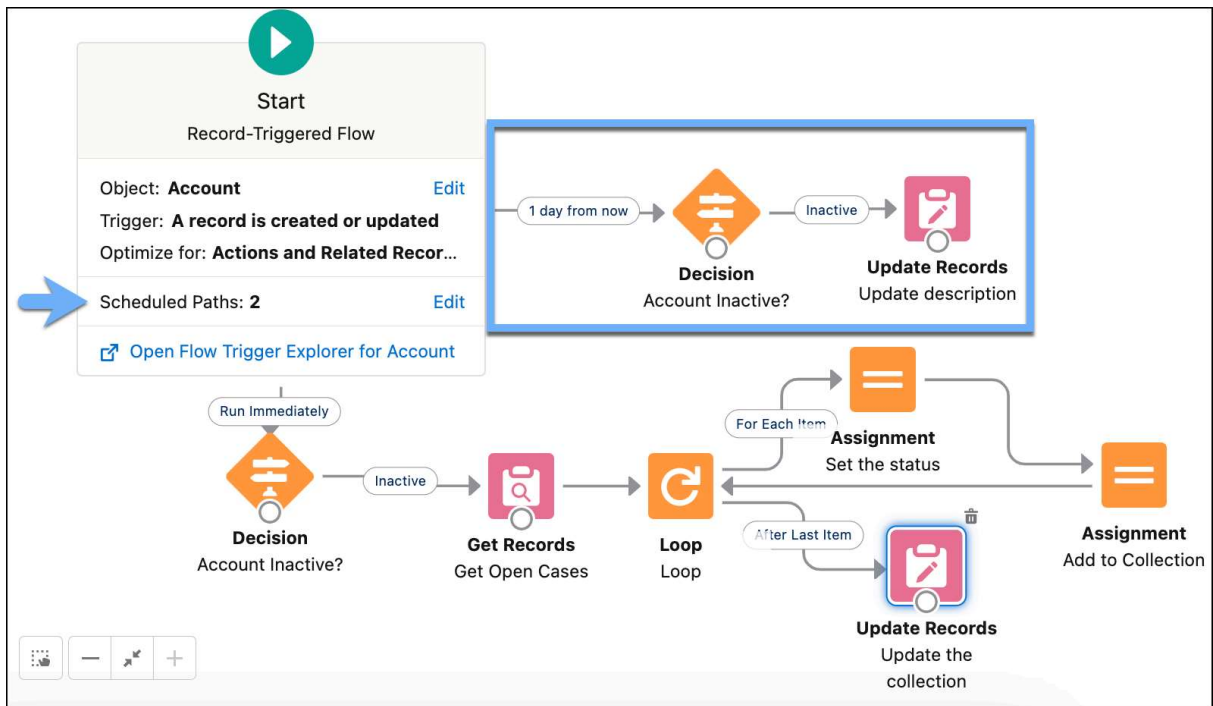
Verify task is auto-created when project status becomes Approved.

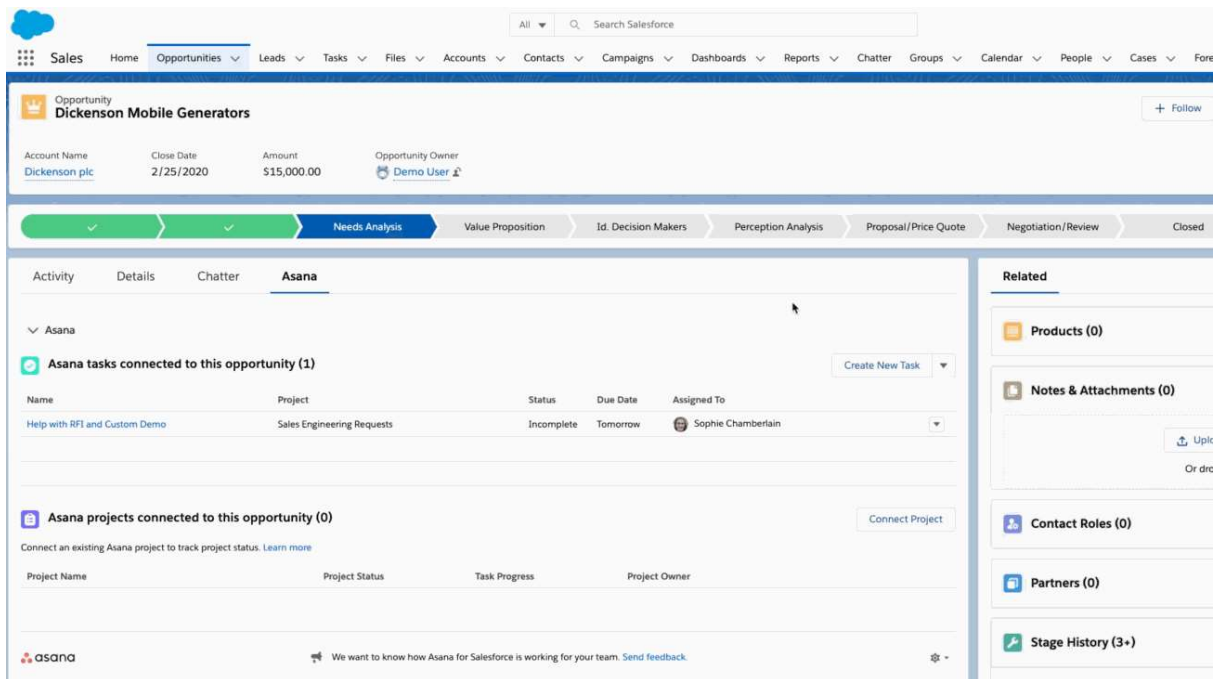
Flowchart

Update Status = Approved

- Flow Triggered
- Task Created







Test Case 4 – Approval Process Testing

Use Case

Verify project moves to Approved after manager approval.

Flowchart

Submit For Approval

→ Manager Approves

→ Status Updated

Business License Application

BLA-0000000002

Account

License Type

Applied Date

Application Status

Salon Establishment License

4/12/2022, 2:33 AM

Submitted

>

Submitted

Returned

In Review

Inspection Phase

Violations Found

Inspection Completed

De

Related

Details

Unified View

Fees

Files (1)

Add Files

Scan 26-Nov-2020

PDF

Jun 17, 2022 • 568KB • pdf

View All

Document Checklist Items (1)

New

Name

Document Type

Status

Created Date

Business Establishment documents

address_proof

New

6/16/2022, 3:50 AM

View All

Travel App

Chatter

Reports

Dashboards

Departments

Travel Approvals

TA-00001

+ Follow

Edit

New Contact

New Opportunity

Related

Details

Travel Approval #

TA-00001

Owner

Flávia Andréa Paolola

Status

Draft

Department

Technology

Total Expenses

\$1,440.00

Status Indicator

Activity

Chatter

Filters: All time • All ac

Refresh

Upcoming & Overdue

No activities to sh

Get started by sending an email, sched

New Case

New Lead

Delete

Clone

Change Owner

Printable View

Sharing

Sharing Hierarchy

Submit for Approval

Lightning Sales Home Opportunities Accounts Contacts Leads TestObj1 Tasks Files Notes Dashboards Reports More

Opportunity TestOpp + Follow Proposal Edit New Opportunity Product

No past activity. Past meetings and tasks marked as done show up here.

Approval History Submit for Approval

Price Confirmation by VP

Step Name	Date	Status	Assigned To	Actual Approver	Comments
VP Approval for Price	2020/05/01 01:59	Approved	VP VP	VP VP	Ok
Submitted	2020/05/01 01:57	Process Started	Sales User	Sales User	please review

Proposal Approval by Manager

Step Name	Date	Status	Assigned To	Actual Approver	Comments
Manager Approval	2020/05/01 01:38	Approved	Mgr1 Mgr1	Mgr1 Mgr1	Seems OK. Go ahead!
Manager Approval	2020/05/01 01:38	No Response	Mgr2 Mgr2	Mgr1 Mgr1	
Submitted	2020/05/01 01:37	Process Started	Sales User	Sales User	Pls review revised version.

Proposal Approval by Manager

Step Name	Date	Status	Assigned To	Actual Approver	Comments
Manager Approval	2020/05/01 01:36	Rejected	Mgr1 Mgr1	Mgr1 Mgr1	Needs to be refined further.
Manager Approval	2020/05/01 01:36	No Response	Mgr2 Mgr2	Mgr1 Mgr1	
Submitted	2020/05/01 01:36	Process Started	Sales User	Sales User	Please review my proposal.

Notes (0)

Products (0)

Contact Roles (0)

Approval History (3+)

VP Approval for Price

Date: 5/1/2020 1:59 AM
Status: Approved
Assigned To: VP VP
Actual Approver: VP VP
Comments: Ok

Approval Request Submitted

Date: 5/1/2020 1:57 AM
Status: Submitted
Assigned To: Sales User
Actual Approver: Sales User
Comments: please review

Manager Approval

Date: 5/1/2020 1:38 AM
Status: Approved
Assigned To: Mgr1 Mgr1
Actual Approver: Mgr1 Mgr1
Comments: Seems OK. Go ahead!

orgLimits departmentMessage FileViewer 便利リンク Macros Notes History

Approval Request SA Campaign: INI-2719 Help for this Page

Back to SA Campaign: INI-2719

Approve/Reject Approval Request

Initiative Code: INI-2719
Initiative/Campaign Name: Lahar05043 金尚东
Initiative Step: Initiative Summary
Number of Programs Allowed: 3
Primary Organization Unit: Cardiovascular and IM Business Unit
SG AHM Initiative Approval Process: Medical Education Packaged Initiative
Submitted Budget: CNY 0.00
Owner: JIN Xiaodong
Comments:

Approve Reject Cancel

Approval History Approval History Help

Date	Status	Assigned To	Actual Approver	Comments	Overall Status
Step: FC Approver Approval (Pending for first approval)					
2018-4-5 PM5:29	Pending	FU Yi	FU Yi		Pending
Step: TA Approvers Approval					
2018-4-5 PM5:29		WANG Bellei			Approved
2018-4-5 PM5:29		LI Ye			
2018-4-5 PM5:29		JackieMAO Yeqing			
2018-4-5 PM5:29		ZHENG Lian			
2018-4-5 PM5:29		IleneZHANG Xi			

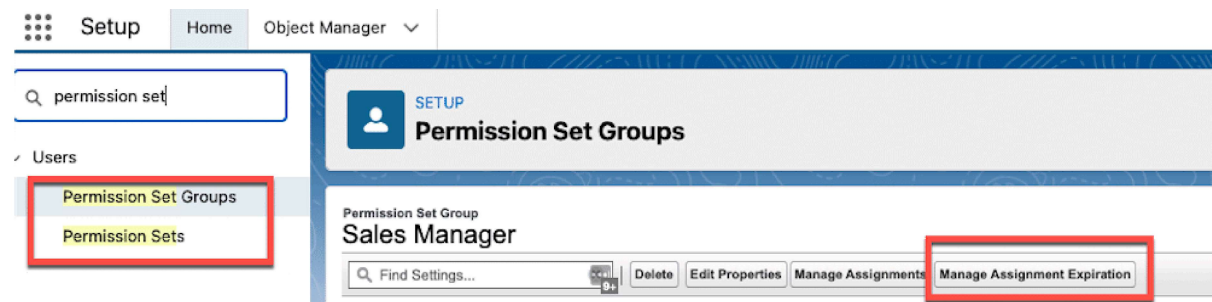
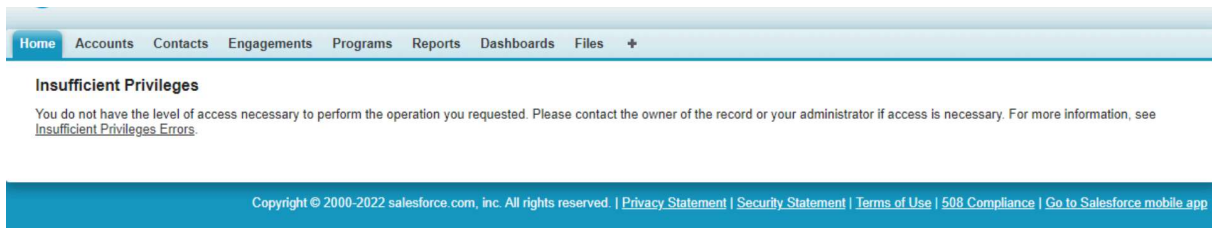
Test Case 5 – Security Testing (Permission Set)

Use Case

Verify user without permission cannot delete project.

Flowchart

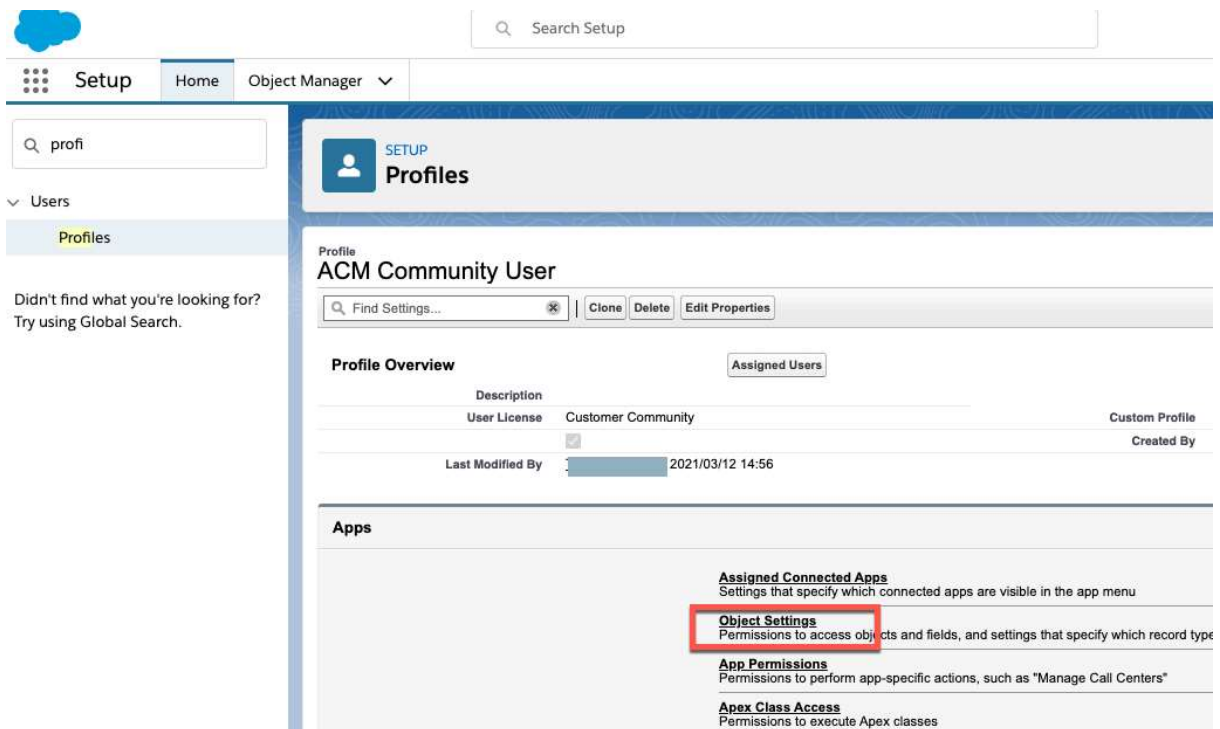
User Attempts Delete
→ Permission Checked
→ Access Denied



What Should Be in a Perm Set vs. a Profile

We are still on the path to End of Life permissions on profiles

Permission Sets	Profiles
System permissions	Use the Minimum Access profile
Object & field permissions	Defaults (Record Types, Apps)
Connected app access	Page Layout Assignment
Apex Classes & VF pages	Login Hours/IP Ranges
Record Types (not defaults)	
Apps	
Tab settings	
Custom permissions	



Test Case 6 – Integration Testing (Callout)

Use Case

Verify data is sent to external system using Named Credential.

Flowchart

Trigger Callout

- Named Credential
- External System
- Response Received



SETUP > NAMED CREDENTIALS

Sample External Credential

[Edit](#)[Delete](#)

Label

Sample External Credential

Name

Sample_External_Credential

Authentication Protocol

OAuth 2.0

Authentication Flow Type

Browser Flow

Scope

Authentication Provider

salesforce@@@@

Managed Package Access

Created By Namespace ⓘ

Related Named Credentials

Label	Name	URL
My Named Credential	My_Named_Credential	https://my_example_endpoint.com

Principals

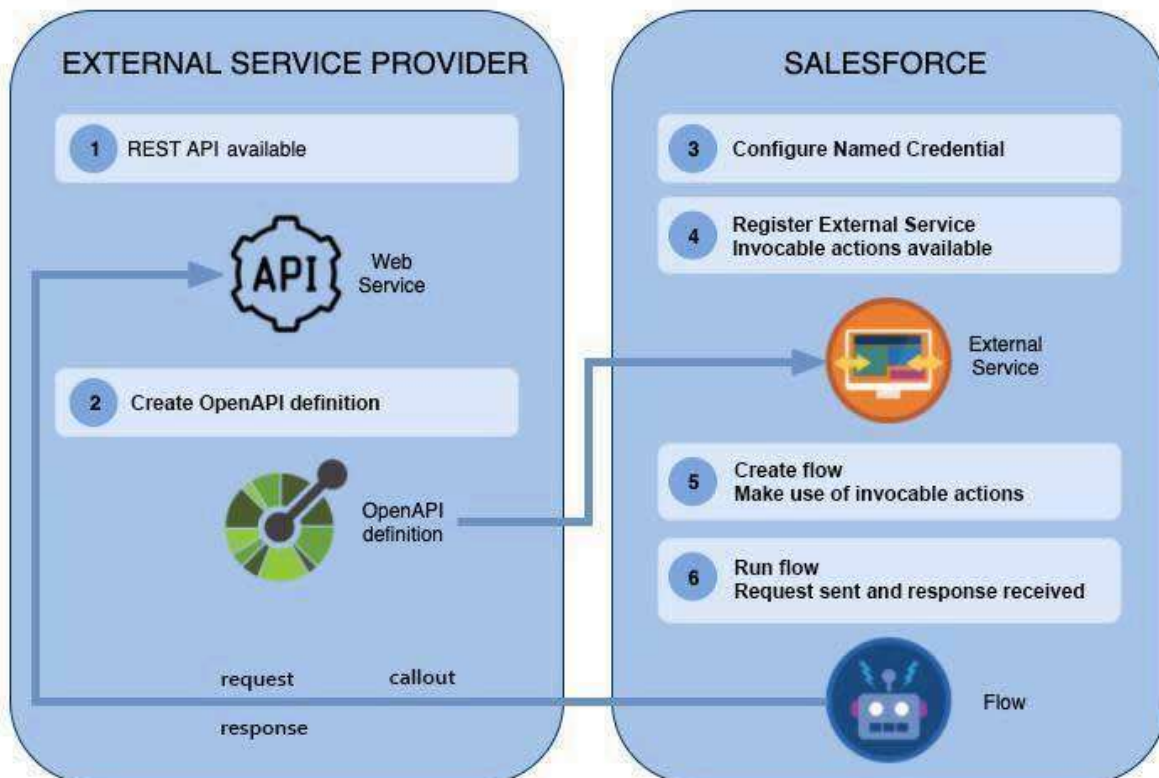
[New](#)

Seq...	Parameter Name	Identity Type	S	Authentic...	Actions
1	PermSetOne@@@	Per User Principal		Not Configured	▼
500	PermSetTwo####	Named Principal		Not Configured	▼
1,000	SamplePermissionSet\$\$\$\$	Named Principal		Not Configured	▼

```

File • Edit • Debug • Test • Workspace • Help •
Animals.apex • AnimalsCalloutMockApex • AnimalsTestApex •
Code Coverage: All Tests 100% • API Version: 53
1 • public class Animals{
2 •     public static HttpResponse animals(){
3         Http http = new Http();
4         HttpRequest req = new HttpRequest();
5         req.setEndpoint('https://th-apex-http-callout.herokuapp.com/animals');
6         req.setMethod('POST');
7         req.setHeader('Content-Type', 'application/json;charset=UTF-8');
8         req.setBody('{\"Name\":\"Lion\"}');
9
10        HttpResponse res = http.send(req);
11
12        if(res.getStatusCode() == 200){
13            System.debug(res.getBody());
14        }
15        else{
16            System.debug('Error, Status :'+ res.getStatusCode());
17        }
18        return res;
19    }

```





Conclusion

The Project Management CRM application was thoroughly tested using multiple scenarios. All major functionalities including record creation, automation, approvals, integrations, and security performed as expected. The system is stable, secure, and ready for deployment. Future enhancements may include AI-based recommendations, chatbot integration, and advanced analytics dashboards.